English

(1001CJA106216230184)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
01-05-2024

JEE(Main+Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\in_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

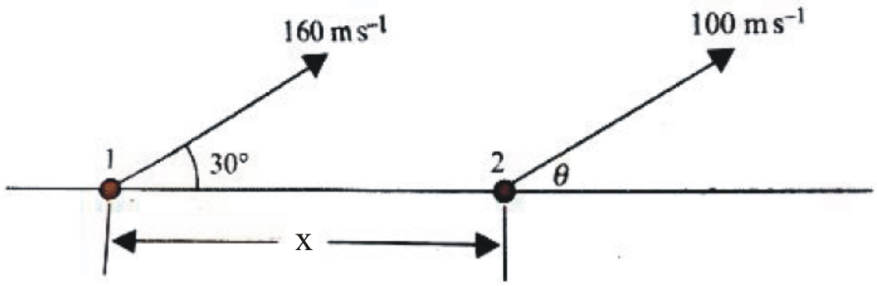
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. Suppose two particles 1 and 2 are projected in vertical plane simultaneously.



Their angles of projection are 30° and θ , respectively, with the horizontal. Let they collide after a time t in air. Then (neglect air resistance)

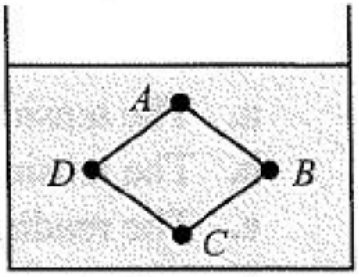
(A) $\theta = \sin^{-1}(4/5)$ and they will have same speed just before the collision.

(B) $\theta = \sin^{-1}(4/5)$ and they will have different speed just before the collision.

(C) $x < 1280\sqrt{3} - 960$m

(D) It is possible that the particles collide when both of them are at their highest point.

2. Figure shows a container filled with a liquid of density ρ . Four points A, B, C and D lie on the vertices of a vertical square. Points A and C lie on a vertical line and points B and D lies on a horizontal line. Choose the correct statement(s) about the pressure at the four points.



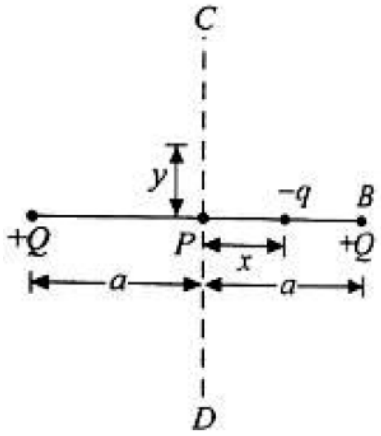
(A) $P_D = P_B$

(B) $P_A < P_B = P_D < P_C$

(C) $P_D = P_B = \frac{P_C - P_A}{2}$

(D) $P_D = P_B = \frac{P_C + P_A}{2}$

3. In the arrangement shown in figure, two positive charges, +Q each, are fixed. Mark the correct statement(s) regarding a third charged particle −q placed at the midpoint P that can be displaced along or perpendicular to the line connecting the charges.



(A) The particle will perform SHM for x << a.

(B) The particle will oscillate about P but not harmonically for any x.

(C) The particle will perform SHM for y << a

(D) The particle will oscillate about P but not harmonically for y comparable to a.

4. A charged particle P leaves the origin with speed $v = v_0$ at some inclination with the x-axis. There is a uniform magnetic field B along the x-axis. P strikes a fixed target T on the x-axis for a minimum value of $B = B_0$. P will also strike T if.

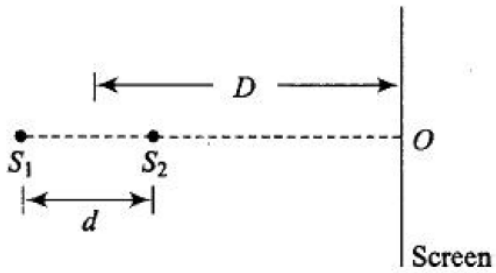
(A) $B = 2B_0, v = 2v_0$

(B) $B = 2B_0, v = v_0$

(C) $B = B_0, v = 2v_0$

(D) $B = \frac{B_0}{2}, v = 2v_0$

5. Two points monochromatic and coherent sources of light of wavelength λ each are placed as shown in figure. The initial phase difference between the sources is zero. (D >> d). Mark the correct statement(s).



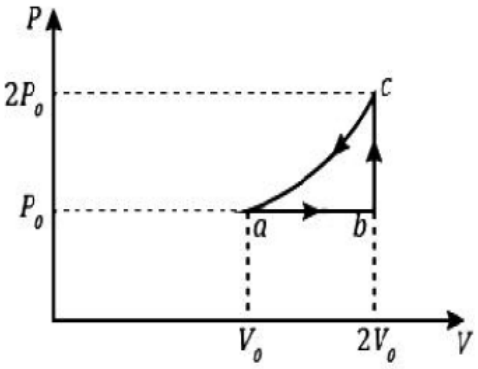
(A) If $d = \frac{7\lambda}{2}$, O will be a minima.

(B) If $d = \lambda$, only one maxima can be observed on the screen.

(C) If $d = 4.8\lambda$, then total 5 minima would be there on the screen.

(D) If $d = \frac{5\lambda}{2}$, the intensity at O would be minimum.

6. One mole of an ideal monatomic gas has initial temperature T_0, is made to go through the cycle abca shown in the given figure. If U denotes the internal energy, then choose the correct alternative.



(A) $U_c > U_b > U_a$

(B) $U_c - U_b = 3RT_0$

(C) $U_c - U_a = \frac{9RT_0}{2}$

(D) $U_b - U_a = \frac{3RT_0}{2}$

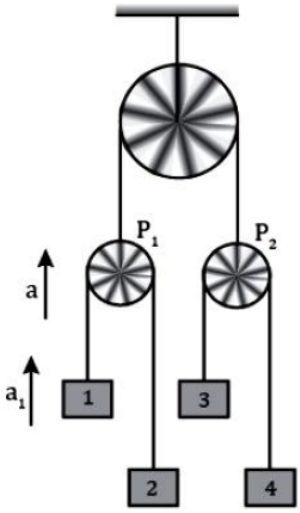
SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

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7. In the system shown in figure, masses of the blocks are such that when system is released, the acceleration of pulley P_1 is a upwards and the acceleration of block 1 is a_1 upward. It is found that the acceleration of block 3 is same as that of 1 both in magnitude and direction. Given that $a_1 > a > a_1/2$. Match the following.



List-I		List-II	
(P)	Acceleration of 2	(1)	$2a + a_1$
(Q)	Acceleration of 4	(2)	$2a - a_1$
(R)	Acceleration 2 w.r.t. 3	(3)	Upward
(S)	Acceleration 2 w.r.t. 4	(4)	Downward

(A) P ⟶ 2,3;Q ⟶ 1,4;R ⟶ 4;S ⟶ 3

(B) P ⟶ 2,3;Q ⟶ 1,2;R ⟶ 4;S ⟶ 3

(C) P ⟶ 2;Q ⟶ 1,4;R ⟶ 3;S ⟶ 4

(D) P ⟶ 3;Q ⟶ 1;R ⟶ 4;S ⟶ 3

8. Bodies A and B are rolling without slipping on a rough inclined plane, starting from rest.

Match the following.

List -I		List -II	
(P)	Velocity of A and B after travelling the same distance will be	(1)	Different if A and B have same size but different shape and mass
(Q)	Time taken by A and B to reach the bottom of the plane from the same height will be	(2)	Different if A and B have same size and same mass but having different shapes
(R)	Acceleration of centre of mass of A and B will be	(3)	Same if A and B have same shape and same size but of different materials
(S)	During rolling motion, frictional force experienced by A and B will be	(4)	Same if A and B have same size but different shape and mass

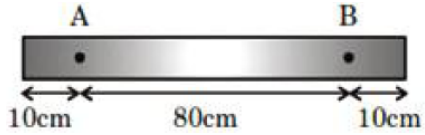
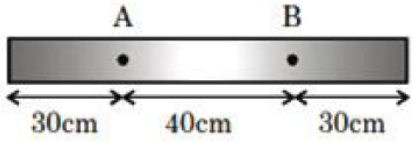
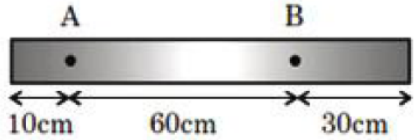
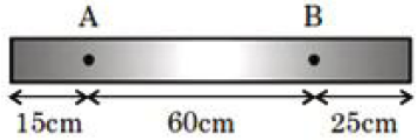
(A) P ⟶ 1,2,3;Q ⟶ 2,3,4;R ⟶ 1,2,3;S ⟶ 1,2

(B) P ⟶ 1,2,3;Q ⟶ 1,2,3;R ⟶ 1,2;S ⟶ 1,2,4

(C) P ⟶ 1,2,3;Q ⟶ 2,3,4;R ⟶ 1,2;S ⟶ 1,2

(D) P ⟶ 1,2,3;Q ⟶ 1,2,3;R ⟶ 1,2,3;S ⟶ 1,2

9. List-I represents the longitudinal vibration of metallic rods each of length 1m and each rod is rigidly clamped at two points A and B. List-II represents their frequency and wavelength of longitudinal vibration in the fundamental mode of vibration. Then match the option of List-I correctly with the options of List-II.

List -I		List -II	
(P)	 $Y = 2 \times 10^{11} N/m^2$, $\rho = 8000\ kg/m^3$	(1)	f = 40 kHz
(Q)	 $Y = 1.6 \times 10^{11} N/m^2$, $\rho = 2500\ kg/m^3$	(2)	f = 12.5 kHz
(R)	 $Y = 2 \times 10^{11} N/m^2$, $\rho = 8000\ kg/m^3$	(3)	f = 20 kHz
(S)	 $Y = 1.6 \times 10^{11} N/m^2$, $\rho = 2500\ kg/m^3$	(4)	$\lambda = 0.4$ m
		(5)	$\lambda = 0.2$ m

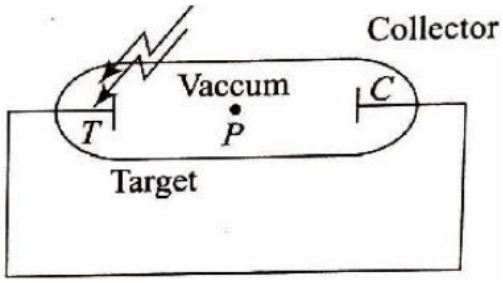
(A) P ⟶ 1,5;Q ⟶ 3,4;R ⟶ 2,5;S ⟶ 1,5

(B) P ⟶ 2,5;Q ⟶ 3,5;R ⟶ 2,4;S ⟶ 2,5

(C) P ⟶ 3,4;Q ⟶ 3,5;R ⟶ 2,4;S ⟶ 2,5

(D) P ⟶ 2,4;Q ⟶ 3,4;R ⟶ 2,4;S ⟶ 1,5

10. In a photoelectric experimental arrangement, light of frequency f is incident on a metal target whose work function is $\phi = \frac{hf}{3}$ as shown. In list-I, KE of photoelectron is mentioned at various locations/instants and in list-II, the corresponding values. Match the entries of list-I with the entries of list-II.



List -I		List -II	
(P)	Maximum KE of photoelectrons just after emission from target	(1)	Zero
(Q)	KE of photoelectrons just after emission from target	(2)	hf/3
(R)	KE of photoelectron when they are halfway between the target and collector	(3)	hf/2
(S)	KE of photoelectrons as they reach the collector	(4)	2hf/3

(A) P ⟶ 4;Q ⟶ 1,2,3,4;R ⟶ 2,3,4;S ⟶ 1

(B) P ⟶ 4;Q ⟶ 1,4;R ⟶ 2,3;S ⟶ 1,3,4

(C) P ⟶ 1,4;Q ⟶ 4;R ⟶ 3,4;S ⟶ 1,3,4

(D) P ⟶ 4;Q ⟶ 1,2,3,4;R ⟶ 1,2,3,4;S ⟶ 1,2,3,4

SECTION-II : (Maximum Marks: 24)

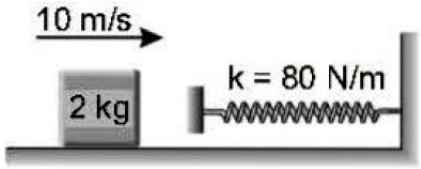
- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

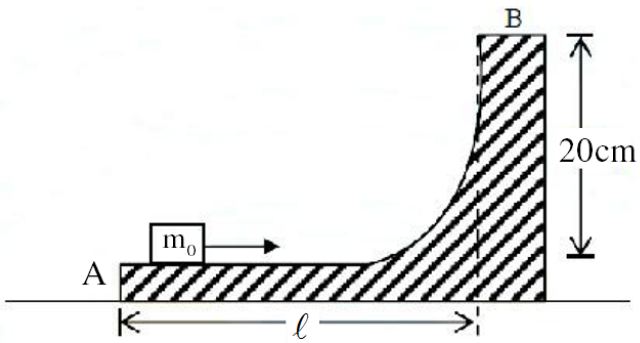
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1. A car travelling at 28 m/s has no tendency to slip (i.e. assume friction force on the car = 0) on the track of radius 200m banked at an angle θ . When the speed is increased to 35 m/s, the car is just on the point of slipping up the track. Calculate the coefficient of friction between the car and the track. (Upto two decimals) ($g = 10 m/s^2$.)

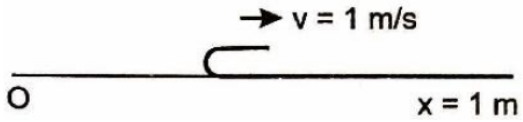
2. A block of mass 2kg approaches a spring of stiffness k = 80 N/m on a smooth horizontal plane with a speed of 10 m/s. Find the time in seconds (upto 2 decimals) in which kinetic energy of the block reduces to 50% of its initial kinetic energy. Assume t = 0 when the block just touches the spring, for the first time.



3. A wedge of mass M=2 m_0 rests on a smooth horizontal plane. A small block of mass m_0 rests over it at left end A as shown in figure. A sharp impulse is applied on the block, due to which it starts moving to the right with velocity v_0 = 6 m/s . At highest point of its trajectory, the block collides with a particle of same mass m_0 moving vertically downwards with velocity v = 2 m/s and gets stuck with it. If the combined mass lands at the end point A of the body of mass M, calculate length ℓ . Neglect friction, take $g = 10 m/s^2$.

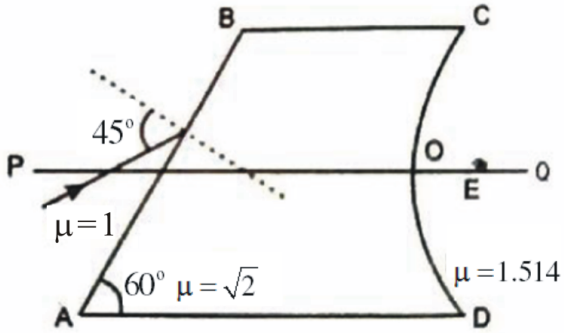


4. A long thin carpet is laid on the floor. One end of the carpet is bent back and then pulled backwards with constant velocity υ = 1 m/s , just above the part of the carpet which is still at rest on the floor. What is the minimum force (in SI system) needed to pull the moving part, if the carpet has unit length and unit mass. (Neglect all dissipative forces) (upto 1 decimal)



5. The volume density of the elastic deformation energy in fresh water at a depth of h = 1 km. (Take density of water (constant) $=10^3$ kg/m^3,g = 9.8 m/s^2 and bulk modulus of elasticity of water B = 2×10^9 N/m^2) is x × 10^4 J/m^3 . Find 'x' (in 2 digit) (neglect atmospheric pressure).

6. Figure shows an irregular block of material of refractive index $\sqrt{2}$. A ray of light strikes the face AB as shown in the figure. After refraction it is incident on a spherical surface CD of radius of curvature 0.4m and enters a medium of refractive index 1.514 to meet PQ at E. Find the distance OE in meters (upto two places of decimal)



7. A photon with an energy of 4.9eV ejects photoelectrons from tungsten. When the ejected electron enters a constant magnetic field of strength B = 2.5 mT at an angle of 60° with the field direction, the maximum pitch of the helix described by the electron is found to be 2.7 mm. Find the work function of the metal in electron volts (upto two digits). Given that specific charge of electron is 1.76×10^{11} C/kg

8. Consider the β – decay $^{198}Au \longrightarrow {}^{198}Hg + \beta^- + \bar{\nu}$
Where ^{198}Hg represents a mercury nucleus in an excited state of energy 1.088 MeV above the ground state. What can be the maximum kinetic energy (MeV) of electron emitted (upto two decimals). The atomic mass of ^{198}Au is 197.968233u and that of ^{198}Hg is 197.966760u. lu = 931 MeV/c^2

PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. Which can form square planar complex with coordination number (CN) is 4?

(A) Pt(II), Pd(II) with strong and weak field ligand

(B) Au(III) with strong and weak field ligand

(C) Ni(II) with C.N. = 4 and strong field ligand

(D) Hg(II) with C.N. = 4 and strong field ligand

2. Froth floatation:

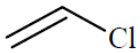
(A) is physical method of separation of mineral from the gangue

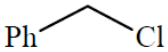
(B) is a method to concentrate the ore depending on the difference in wettability of gangue and the ore

(C) is used for the sulphide ores

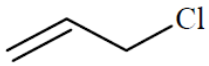
(D) is a method in which impurities sink to the bottom

3. Both S_N1 and S_N2 reactions are not favourable in which of the given compounds?

(A) 

(B) 

(C) Ph–Cl

(D) 

4. Which of the following molecule(s) have higher resonance energy than 1,3–butadiene?

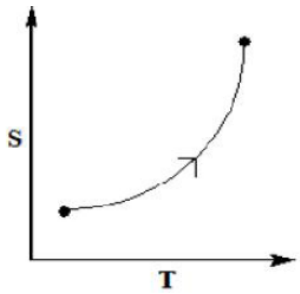
(A) Pyrrole

(B) Thiophene

(C) Tetrahydro Furan

(D) 1,4–cyclohexadiene

5. A mixture of fixed number of moles of ideal gases undergoes a thermodynamic process as shown by the plot. Which of the following need not be same for the process, if the composition of ideal gas mixture changes for the same initial and final temperatures? S= molar entropy of the mixture, T= temperature



(A) Change in internal energy

(B) Change in free energy

(C) Work done

(D) Change in enthalpy

6. Given the data at 25°C

$Ag + I^- \longrightarrow AgI + e^-$; $E^0 = 0.152V$, $Ag \longrightarrow Ag^+ + e^-$; $E^0 = -0.800V$

What is the value of log K_{sp} for AgI? $\left(2.303\frac{RT}{F} = 0.059V\right)$

(A) – 8.12

(B) + 8.612

(C) – 37.83

(D) – 16.13

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

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7. Match the property of the compound given in List–II with the compound given in List–I.

List – I		List– II	
(A)	$KHCO_3$	(P)	contain polymeric anion
(B)	$NaHCO_3$	(Q)	contain dimeric anion
(C)	$LiHCO_3$	(R)	contain hydrogen bond
(D)	$Mg(HCO_3)_2$	(S)	liberates CO_2 on heating
		(T)	can be prepared in solid state

(A) A ⟶ QRST;B ⟶ PRST;C ⟶ S;D ⟶ ST

(B) A ⟶ QRST;B ⟶ PST;C ⟶ PS;D ⟶ RST

(C) A ⟶ QST;B ⟶ PST;C ⟶ S;D ⟶ RST

(D) A ⟶ QST;B ⟶ PRST;C ⟶ PS;D ⟶ ST

8. Match the compound given in List–I with the character given in List–II.

List-I		List-II	
(A)	Hydrogen hypo phosphate ion $H_2P_2O_6^{2-}$	(P)	All hydrogens are ionizable
(B)	Hydrogen pyrophosphate ion $H_2P_2O_7^{2-}$	(Q)	Number of acidic hydrogens present is 2
(C)	Pyrosulphurous acid $H_2S_2O_5$	(R)	sp^3 hybridised central atom
(D)	Hydrogen peroxy mono sulphate ion HSO_5^-	(S)	Overall non–planar structure
		(T)	Can act as oxidizing agent

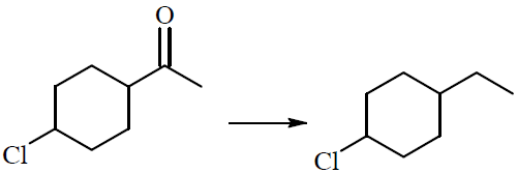
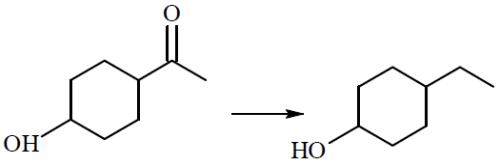
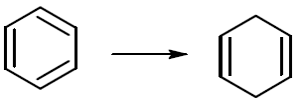
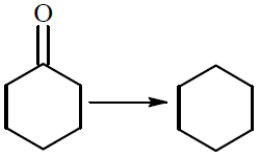
(A) A ⟶ PQRS;B ⟶ RST;C ⟶ PQRS;D ⟶ PQRST

(B) A ⟶ PQS;B ⟶ RST;C ⟶ PRS;D ⟶ PRST

(C) A ⟶ PQS;B ⟶ RS;C ⟶ PRS;D ⟶ PQRST

(D) A ⟶ PQRS;B ⟶ PQRS;C ⟶ PQRS;D ⟶ PRST

9. Match the correct reduction method from List–II for the reaction in List–I.

List-I		List-II	
(A)		(P)	Birch reduction
(B)		(Q)	Stephen's reduction
(C)		(R)	Wolff–Kishner reduction
(D)		(S)	Clemmensen reduction

(A) A ⟶ S;B ⟶ R;C ⟶ P;D ⟶ RS

(B) A ⟶ R;B ⟶ P;C ⟶ RS;D ⟶ S

(C) A ⟶ P;B ⟶ RS;C ⟶ S;D ⟶ R

(D) A ⟶ RS;B ⟶ S;C ⟶ R;D ⟶ P

10. Match the crystal lattice in List–I with the corresponding Geometrical or Physical property in List–II.

List – I (Crystal lattice)		**List – II (Geometrical or Physical property)**	
(A)	Rock salt unit cell	(P)	Three–fold axis of symmetry
(B)	Zinc blende unit cell	(Q)	Four–fold axis of symmetry
(C)	Fluorite unit cell	(R)	Plane of symmetry
(D)	FCC unit cell	(S)	Anisotropy
		(T)	Number of first nearest identical species is 12

(A) A ⟶ PQST;B ⟶ PQRS;C ⟶ PRST;D ⟶ PQRST

(B) A ⟶ PQRST;B ⟶ PQRST;C ⟶ PQRS;D ⟶ PQRST

(C) A ⟶ PQRST;B ⟶ QRS;C ⟶ PQRST;D ⟶ PQRST

(D) A ⟶ PQRST;B ⟶ PQRST;C ⟶ PQRT;D ⟶ PQRT

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

---

1. A mixture contains one gram equivalent of each of the following metal hydroxides. Water is added to the mixture, boiled and filtered. How many of the hydroxides remain in the residue?
NaOH, KOH, $Ba(OH)_2$, $TlOH$, $Be(OH)_2$, $Mg(OH)_2$, $Al(OH)_3$, $Sn(OH)_2$, $Zn(OH)_2$, $Sr(OH)_2$.

2. Polyphosphates are the condensed products of tetrahedral phosphate units that contain P – O – P bonds. A sample of poly phosphoric acid was dissolved in water and titrated with dilute NaOH (aq). Two stoichiometric points were observed at 30.4 and 45.6 cm^3. The number of P – O – P bonds in that poly phosphoric acid is

3.  $\xrightarrow{Cl_2/hv}$ C_4H_7Cl $\xrightarrow[distillation]{Fractional}$ Q .
P(number of isomeric products) (number of fractions)
Find the value of P + Q .

4. An organic compound (X) shows following characteristics
i) It is optically active
ii) Gives +ve haloform test
iii) Gives violet colour with neutral $FeCl_3$ solution
Then the minimum possible molecular weight of compound (X) is

5. How many compounds are more reactive than ethene towards electrophilic addition reaction?

(I) $CH_3 - CH = CH_2$

(II) $H-\overset{\overset{O}{\|}}{C}-CH=CH_2$

(III) $Cl - CH = CH_2$

(IV) $CH_3 - O - CH = CH_2$

(V) $H - C \equiv C - H$

(VI) $Ph - CH = CH_2$

(VII) $CH_2 = CH - CH = CH_2$

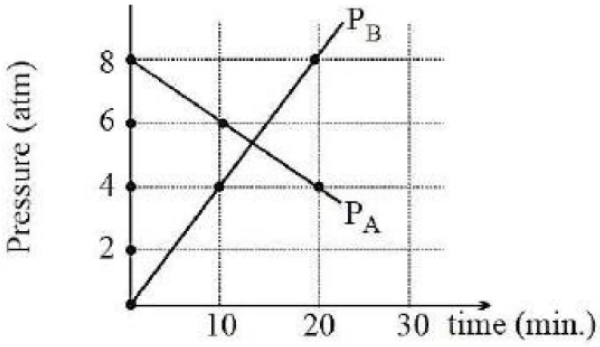
6. 2 mole NO and 16 gm O_2 were taken in a 6.25 L vessel at 27^oC. Calculate the change in pressure when half of the maximum amount of NO_2 has been produced according to reaction.

(Take R = $\frac{1}{12}$ litre.atm/mol K)

$2NO_{(g)} + O_{2(g)} \longrightarrow 2NO_{2(g)}$

7. The energy of the first orbit of hydrogen atom is -13.6 eV. If the fourth ionization enthalpy of gaseous isolated beryllium atom is expressed as $Y \times 10^{-19}$ J, what is the value of 'Y'? Assume nucleus is at rest in either cases. (1 eV = 1.6×10^{-19} J)

8. In a chemical reaction where A(g) is converting to B(g), the following observation is made. What is the time of completion for the reaction (in min)?



PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

1. $f : R-\{0\} \longrightarrow R$ is a differentiable function such that $\int_1^{xy} f(t)dt = y\int_1^{x} f(t)dt + x\int_1^{y} f(t)dt \ \forall\ x, y \in R - \{0\}$, $f(1) = 1$. Which of the following is correct?

(A) $\lim\limits_{x\to\infty} \dfrac{f(x)}{x} = 0$

(B) $\int_{1/e}^{1} f(x)dx = \dfrac{1}{e}$

(C) Number of solutions of $f(x) = f^{-1}(x)$ for $x > 0$ is 1

(D) Number of functions satisfying the given conditions can not be finite

2. For a given hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$. Let P is any point on it such that if a tangent drawn at P is made to meet its conjugate axis in 'T'. PN is the ordinate of point P and PT meets transverse axis in 'T^1', then (A, A^1 are the vertices with positive and negative x -co ordinates and C is its centre).

(A) $CT.PN = 4$

(B) $CN.CT^1 = 9$

(C) $CN.NT^1 = AN.A^1N$

(D) $CN.NT^1 = CN^2 + AC^2$

3. Let a, b, c be integers and $a^2 + b^2 + c^2 = 2$. Let $A = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ consider the system of equations

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Then which of the following statement(s) is/are correct -

(A) The probability that the system of equations has unique solution is 1/2

(B) The probability that the system of equations has infinitely many solutions is 1/2

(C) If $a = 0$, then the number of ordered pairs (b, c) for which the system has no solution is 2

(D) The number of integers in the range of $ab + bc + ca$ is 2

4. The plane containing the line $\frac{x-3}{2} = \frac{y-1}{4} = \frac{z-2}{4}$ is rotated through an angle $\frac{\pi}{2}$ about this line to contain the point (4, 3, 7) then which of the following statements is/are true ?

(A) The equation of plane before rotation is $2x + 4y - 5z = 0$

(B) The equation of plane after rotation is $2x - y - 5 = 0$

(C) The equation of plane before rotation is $x - 2y + 2z - 1 = 0$

(D) The equation of plane after rotation is $x + 2y + 2z = 1$

5. For the equation $|x^2 - a| = \cos^{-1}(\cos x)$ to have some solution the value of 'a' can be

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $-\frac{1}{2}$

(D) $-\frac{1}{4}$

6. For ΔABC, let $A_1B_1C_1$ be the triangle formed by the excentres of ΔABC, let $\Delta A_2B_2C_2$ be formed by the excentres of $\Delta A_1B_1C_1$ and so on. Finally let $\Delta A_nB_nC_n$ be the nth derived triangle that it is formed by the excentres of $\Delta A_{n-1}\, B_{n-1}C_{n-1}$ then

(A) $\angle A_1 = 90^\circ - \frac{A}{2}$

(B) The circumcentre of ΔABC always coincides with the orthocentre of $\Delta A_1B_1C_1$

(C) as $n \longrightarrow \infty$, $\Delta A_nB_nC_n$ tends to become equilateral

(D) $B_1C_1 = a\cot\frac{A}{2}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

7. Let O, O^1 and G be the circumcentre, orthocentre and centroid of ΔABC respectively. Now match the values in list-I and list-II.

List-I		List-II	
(A)	If S is any point in plane of A,B,C then $\dfrac{\left\lvert\overrightarrow{SA}+\overrightarrow{SB}+\overrightarrow{SC}\right\rvert}{\left\lvert\overrightarrow{SG}\right\rvert}=$	(P)	4
(B)	$\dfrac{\left\lvert\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}\right\rvert}{\left\lvert\overrightarrow{OO'}\right\rvert}=$	(Q)	3
(C)	$\dfrac{\left\lvert\overrightarrow{O'A}+\overrightarrow{O'B}+\overrightarrow{O'C}\right\rvert}{\left\lvert\overrightarrow{OO'}\right\rvert}=$	(R)	2
(D)	$\dfrac{\left\lvert\overrightarrow{AO'}+\overrightarrow{O'B}+\overrightarrow{O'C}\right\rvert}{\left\lvert\overrightarrow{AP}\right\rvert}=$	(S)	1
		(T)	1/2

(Where $\left\lvert\overrightarrow{AP}\right\rvert$ is diameter of circumcircle of ΔABC)

(A) A $\longrightarrow$ Q;B $\longrightarrow$ S;C $\longrightarrow$ R;D $\longrightarrow$ S

(B) A $\longrightarrow$ Q;B $\longrightarrow$ R;C $\longrightarrow$ S;D $\longrightarrow$ S

(C) A $\longrightarrow$ S;B $\longrightarrow$ S;C $\longrightarrow$ Q;D $\longrightarrow$ R

(D) A $\longrightarrow$ S;B $\longrightarrow$ Q;C $\longrightarrow$ R;D $\longrightarrow$ R

8.

List-I		List-II	
(A)	If $\bar{a}=\hat{i}+\hat{j}+\hat{k}$, $\bar{b}=\hat{i}-\hat{j}+\hat{k}$, $\bar{c}=\hat{i}+2\hat{j}-\hat{k}$ then the value of $\begin{vmatrix} \bar{a}.\bar{a} & \bar{a}.\bar{b} & \bar{a}.\bar{c} \\ \bar{b}.\bar{a} & \bar{b}.\bar{b} & \bar{b}.\bar{c} \\ \bar{c}.\bar{a} & \bar{c}.\bar{b} & \bar{c}.\bar{c} \end{vmatrix}$ is equal to	(P)	$\sqrt{3}$
(B)	If points P, Q and R have position vectors $\bar{r}_1=3\hat{i}-2\hat{j}-\hat{k}$, $\bar{r}_2=\hat{i}+3\hat{j}+4\hat{k}$ and $\bar{r}_3=2\hat{i}+\hat{j}-2\hat{k}$ respectively relative to an origin O then the distance of P from the plane OQR is	(Q)	28
(C)	A line with direction ratios (2, 1, 2) intersects the lines $\bar{r}=-\hat{j}+\lambda(\hat{i}+\hat{j}+\hat{k})$ and $\bar{r}=-\hat{i}+\mu(2\hat{i}+\hat{j}+\hat{k})$ at A and B then AB is equal to	(R)	16
(D)	In a regular tetrahedron, the centres of four faces are the vertices of a smaller tetrahedron, The ratio of the volume of the smaller tetrahedron to that of the larger is $\frac{m}{n}$ where m and n are relatively prime positive integers. The value of m + n is	(S)	3

(A) A $\longrightarrow$ R;B $\longrightarrow$ Q;C $\longrightarrow$ S;D $\longrightarrow$ Q

(B) A $\longrightarrow$ R;B $\longrightarrow$ S;C $\longrightarrow$ S;D $\longrightarrow$ Q

(C) A $\longrightarrow$ S;B $\longrightarrow$ Q;C $\longrightarrow$ S;D $\longrightarrow$ Q

(D) A $\longrightarrow$ Q;B $\longrightarrow$ R;C $\longrightarrow$ S;D $\longrightarrow$ Q

9. Let $f(x)=\min\{\sec^2\pi x, 3-\sqrt{2}\sin\pi x\}$, $x \in (0, \frac{1}{2})$

List-I		List-II	
(A)	Number of integers in the range of $f(x)$ is	(P)	0
(B)	Number of points of local extremum of $f(x)$ is	(Q)	2
(C)	Number of values of x where $f(x)$ is an integer is	(R)	1
(D)	Number of solutions of the equation $f(x) = 1$ is	(S)	3
		(T)	4

(A) A $\longrightarrow$ S;B $\longrightarrow$ R;C $\longrightarrow$ Q;D $\longrightarrow$ P

(B) A $\longrightarrow$ R;B $\longrightarrow$ Q;C $\longrightarrow$ Q;D $\longrightarrow$ S

(C) A $\longrightarrow$ R;B $\longrightarrow$ R;C $\longrightarrow$ R;D $\longrightarrow$ R

(D) A $\longrightarrow$ P;B $\longrightarrow$ S;C $\longrightarrow$ Q;D $\longrightarrow$ R

10. Match the following

List-I		List-II	
(A)	$\sqrt{\sin x} : \left[0, \frac{\pi}{3}\right] \to [0, 1]$	(P)	one-one
(B)	$\frac{x+3}{x-1} : (1, \infty) \to (1, \infty)$	(Q)	many-one
(C)	$\sin x : \left[-\frac{\pi}{2}, \frac{4\pi}{3}\right] \to [-1, 1]$	(R)	into
(D)	$\frac{x^2}{x-2} : (2, \infty) \to [8, \infty)$	(S)	onto

(A) A $\longrightarrow$ P,S;B $\longrightarrow$ P,R;C $\longrightarrow$ Q,S;D $\longrightarrow$ Q,S

(B) A $\longrightarrow$ Q,R;B $\longrightarrow$ P,S;C $\longrightarrow$ P,S;D $\longrightarrow$ Q,S

(C) A $\longrightarrow$ P,R;B $\longrightarrow$ P,S;C $\longrightarrow$ Q,S;D $\longrightarrow$ Q,S

(D) A $\longrightarrow$ P,R;B $\longrightarrow$ P,Q;C $\longrightarrow$ Q,S;D $\longrightarrow$ Q,S

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

---

1. If Z is a non-real complex number such that $\arg\left(z^{1/3}\right) = \frac{1}{2}\arg\left(z^2 + \bar{z}z^{1/3}\right)$ then the value of $|z|$ is

2. A circle concentric to the ellipse $\frac{4x^2}{289} + \frac{y^2}{\lambda^2} = 1\left(\lambda < \frac{17}{2}\right)$ passes through foci S_1, S_2 and cuts ellipse at point P. If area of ΔPS_1S_2 is 30 sq.units then S_1S_2 =

3. Let a, b, c are roots of $x^3 + x^2 - 2x + 1 = 0$ then the value of $\begin{vmatrix} a+c & b & a+2b+2c \\ a+b & c & b+2a+2c \\ b+c & a & c+2a+2b \end{vmatrix}$ is equal to

4. If $a_1 = 1$ and $a_n = n(1 + a_{n-1})\ \forall\ n \geq 2$ then $\left(\lim_{n\to\infty}\left(1+\frac{1}{a_1}\right)\left(1+\frac{1}{a_2}\right).....\left(1+\frac{1}{a_n}\right)\right).\frac{1}{e}$ =

5. Let $f(x) = x^2 + 6x + 1$ and S denotes the set of points (x, y) in the coordinate plane such that $f(x) + f(y) \leq 0$ and $f(x) - f(y) \leq 0$. The area of S is $k\pi$ then k =

6. Let p, $q^2 \in Q$, the set of rational numbers and $\cos^2\frac{\pi}{8}$ be a root of the equation $x^2 + px + q^2 = 0$ then the value of $[|\cos p| + |\sin p|] + p + 2\,|\log_2|q||$ equals (Where[.] is G.I.F.)

7. Let $x, y \in R$ such that $\cos x \cos y + 2\sin y + 2\sin x \cos y = 3$ then the value of $\tan^2 x + 5\tan^2 y$ is

8. If $\lim_{n\to\infty}\left(\frac{{}^{3n}c_n}{{}^{2n}c_n}\right)^{\frac{1}{n}} = \frac{a}{b}$ where a, b are relatively prime positive integers then $\frac{a+b+1}{a-b}$ =

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them.	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.
_____	_____
Signature of the Candidate	Signature of the Invigilator

Space for Rough Work



English

(1001CJA106216230185)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
01-05-2024

JEE(Main+Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\in_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

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1. At the initial moment three point A, B and C are on a horizontal plane along a straight line such that AB = BC. Point A begins to move upward with a constant velocity 'v' and point C downward without any initial velocity at a constant acceleration 'a'. If the points being to move simultaneously, then the initial velocity and acceleration of point B for all the three particles to be constantly on same straight line must be

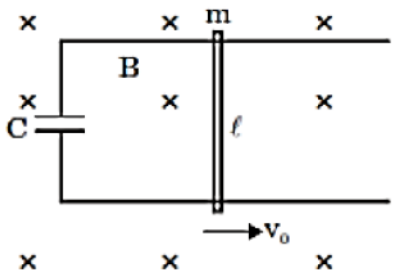
(A) $\frac{v}{2}$ upwards, $\frac{a}{2}$ downwards

(B) $\frac{v}{2}$ upwards, $\frac{a}{2}$ upwards

(C) $\frac{v}{2}$ downwards, $\frac{a}{2}$ downwards

(D) $\frac{v}{2}$ downwards, $\frac{a}{2}$ upwards

2. A wire (with some resistance) of mass m & length ℓ can slide freely on a pair of parallel smooth horizontal rails. A uniform magnetic field B exist in the region in the direction perpendicular to the plane of the rails. The rails are connected at left end by a capacitor of capacity C. At t = 0, wire is given a velocity v_0 towards right side. Then



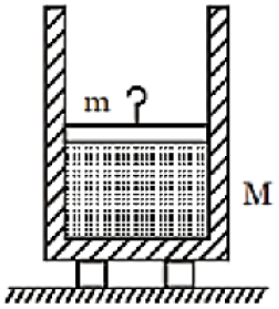
(A) Wire stops finally

(B) Wire will oscillate

(C) Wire will move with constant velocity after sufficient time.

(D) Final K.E. of the rod must be equal to the energy stored in the capacitor

3. As shown, a piston chamber of cross section area A is filled with ideal gas. A sealed piston of mass m is right at the middle height of the cylinder at equilibrium. The friction force between the chamber wall and the piston can be ignored. The mass of the gas + chamber is M. The atmosphere pressure is P_0 . Now slowly pull the piston upwards, find the maximum value of M such that the chamber can be lifted off the ground. The temperature remains unchanged.



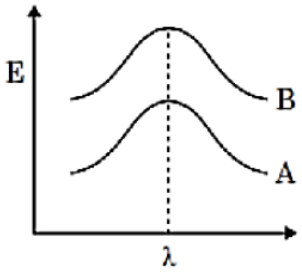
(A) $M = \dfrac{P_0A - 2mg}{2g}$

(B) $M = \dfrac{P_0A - mg}{2g}$

(C) $M = \dfrac{P_0A - mg}{g}$

(D) $M = \dfrac{P_0A}{2g}$

4. Two bodies A and B have the same surface area and mass. The bodies have absolute temperature T_A and T_B , emissivities e_A and e_B , and specific heat capacities S_A and S_B . The intensity of radiation (E) is plotted against the wavelength (λ) of radiation for both bodies. Which of the following is possible?



(A) $T_A = T_B$, $e_A = e_B$

(B) $T_A = T_B$, $e_A \neq e_B$

(C) $T_A \neq T_B$, $e_A = e_B$,

(D) $T_A \neq T_B$, $e_A \neq e_B$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. The pitch of screw gauge is 1 mm, and its circular scale is divided into 100 divisions. When nothing is put between the studs the zero of main scale is not seen but when circular scale is rotated by 450° the zero of main scale is just visible and the zero of main scale coincides with the zero of circular scale. When a glass plate is placed between the studs, the circular scale lies between 18th and 19th division of main scale and circular scale reads 34 divisions. Then

(A) There is a positive zero error, and its magnitude is 1.25 mm.

(B) There is a negative zero error, and its magnitude is 1.25 mm

(C) The thickness of the glass plate is 19.59 mm.

(D) The thickness of the glass plate is 17.09 mm

6. A transverse harmonic wave is propagating along a taut string. Tension in the string is 50 N and it's linear mass density is 0.02 kg/m. The string is driven by a 80 Hz oscillator tied to one end, oscillating with amplitude of 1 mm. The other end of string is terminated so that all the wave energy is absorbed and there is no reflection. From the given data, choose the **correct** option(s) (take $\pi^2 = 10$)

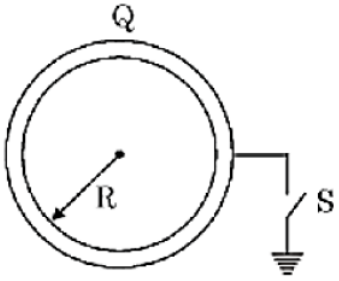
(A) Power of the oscillator is 0.128 W

(B) Average energy of the wave on a 1 m long segment of the string is 2.56 mJ

(C) If tension is the string is quadrupled and power of oscillator remain same, then new amplitude of oscillation of string is $\frac{1}{\sqrt{2}}$ mm

(D) Power of oscillator is 0.252 W

7. An isolated metallic spherical thin shell has given a charge Q as shown. The parameter r is the distance to be measured from center of sphere. Which of the following statement(s) is/are **CORRECT**.



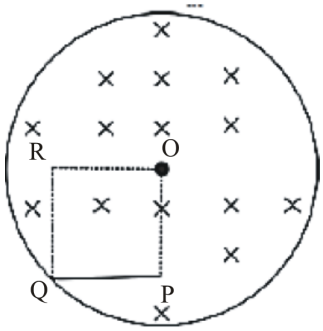
(A) The energy stored in space $R < r < 2R$ is $\frac{Q^2}{16\pi\varepsilon_0 R}$

(B) The energy stored in space $R < r < 2R$ is $\frac{Q^2}{32\pi\varepsilon_0 R}$

(C) Energy stored in space $0 < r < \frac{R}{2}$ is $\frac{Q^2}{16\pi\varepsilon_0 R}$

(D) If switch S is closed the heat generated in wire is $\frac{Q^2}{8\pi\varepsilon_0 R}$

8. Magnetic field exist in a cylindrical region of radius 'R' and centre 'O' as shown in figure. A conducting rod PQ of length 'a' is kept in the field as shown in figure. OPQR forms a square of side 'a'. Magnetic field in the region is changing at the rate $\frac{dB}{dt} = \alpha(\alpha > 0)$ then



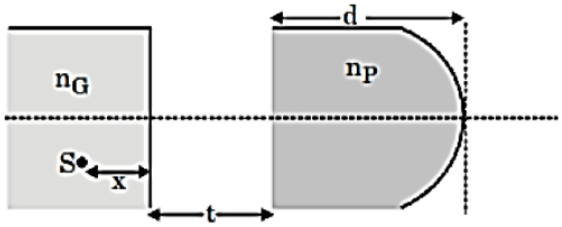
(A) Line integral $\int_P^Q \vec{E}_{ind}.\vec{dl} = \frac{-a^2\alpha}{2}$

(B) Potential difference between P & Q is $\frac{a^2\alpha}{2}$

(C) Line integral $\int_P^Q \vec{E}_{ind}.\vec{dl} = \frac{a^2\alpha}{2}$

(D) Potential difference between Q & R is $\frac{a^2\alpha}{2}$

9. As shown below, a point light source (S) is inside a glass block of refractive index $n_G = \frac{3}{2}$, at a distance 2 mm from the optical axis and distance x = 3 cm from the flat surface. A thick plano-convex lens made of transparent plastic material of refractive index $n_P = 2$, is placed near the source as shown. The thickness of the lens is d, and the radius of convex surface is R = 15 cm. The width of air gap between the glass block and the lens is 't'. The width of air gap t and thickness of lens d is adjusted so that light (paraxial rays) from point source forms always a parallel beam after passing through the lens. Select **CORRET** option(s).



(A) For d = 16cm, width of air gap t = 5 cm

(B) For thickness d = 10 cm, width of air gap t = 8 cm

(C) The maximum adjustable value of d = 26 cm.

(D) The maximum adjustable value of d = 16 cm.

10. $A \longrightarrow B + C$

A radioactive nucleus A at rest disintegrates into two nuclei B and C. Mass of B is 12 m and that of C is 4m. The Q-value of the reaction is, $Q = \frac{h^2}{24m\lambda^2}$. The energy liberated in reaction is completely imparted to the products (B and C) as kinetic energy.

(A) The de Broglie wavelength of B is 2λ

(B) The de Broglie wavelength of C is 2λ

(C) The mass of the nucleus A is $16m + \frac{h^2}{24mc^2\lambda^2}$

(D) The de Broglie wavelength of C is $\frac{2\lambda}{3}$

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

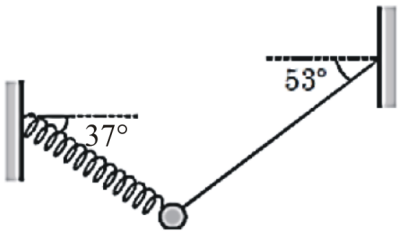
- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If only the correct answer is given.
Zero Marks	:	0	If no answer is given.
Negative Marks	:	–1	In all other cases

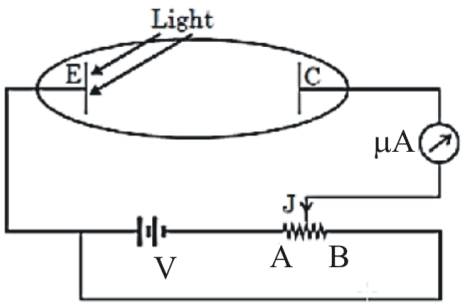
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1. A horizontal stretched string, fixed at two ends, is vibrating in its 39th overtone according to the equation, $y(x,t)=(0.01m)\sin[(62.8m^{-1})x]\cos[(628s^{-1})t]$. Assuming $\pi = 3.14$, what is the length of string (in m)?

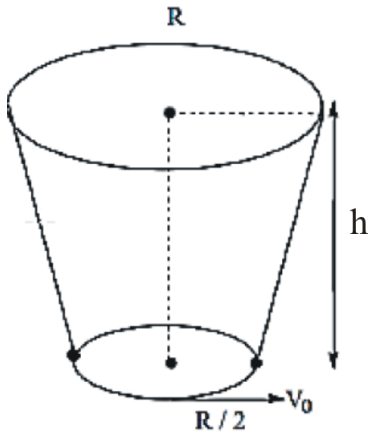
2. A body of mass 1 kg is suspended by means of a spring and a thread between two walls as shown in the figure. The measure of both angles between the thread and the horizontal and the spring and the horizontal is shown in the figure. What is the initial acceleration (in m/ s^2) immediately after the string is cut (use g = 10 m/s^2)



3. In a photoelectric experimental setup ultraviolet light of wavelength 350 nm is incident on emitter plate (E). The work function of the emitter plate is ϕ = 2.2eV. AB is a uniform wire resistor having length L = 100 cm. Resistance of the wire AB is 10 Ω and the emf of the battery is V = 10 volt. The sliding contact J can be moved along the wire AB. When the sliding contact is placed at end B of the wire resistor the micro-ammeter shows a reading of i = ±6μA. Assume that photoelectric current is very small compared to the current through the cell. Find the reading of the ammeter (in μA) when the slider is moved to end A of the wire.

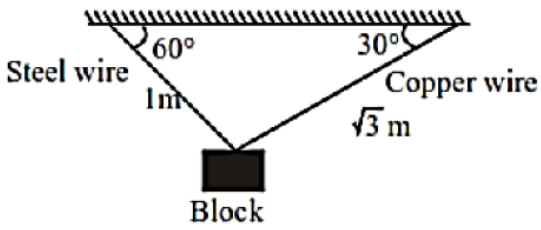


4. A smooth fixed cylindrical glass lies on the ground. A small particle of mass m lies on the bottom surface touching circumference as shown in the figure. The minimum value of V_0 (along the circumference) such that particle just reaches at the top of glass is $\sqrt{\frac{Kgh}{3}}$. Find the value of K.



5. A block of weight 100 N is suspended by copper and steel wires of same cross sectional area 0.5 cm^2 and, length $\sqrt{3}$m and 1 m, respectively. Their other ends are fixed on a ceiling as shown in figure. The angles subtended by copper and steel wires with ceiling are 30° and 60° , respectively. If elongation in copper wire is $(\Delta \ell_C)$ and elongation in steel wire is $(\Delta \ell_S)$, then the ratio $\frac{\Delta \ell_C}{\Delta \ell_S}$ is____.

[Young's modulus for copper and steel are $1 \times 10^{11} N/m^2$ and $2 \times 10^{11} N/m^2$ respectively]

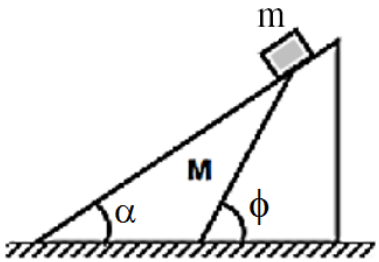


6. A water channel of width b = 17 m lies between two frictionless platforms A and B. A plank (P) of length ℓ = 2 m rest near the platform A. A block of mass m = 25 kg sliding on the platform A lands on the plank and stays on it. Force $\vec{F}$ of water resistance on the plank is proportional to velocity of the plank $(\vec{v}_{pr})$ relative to the water and given by the law $\vec{F} = -k\vec{v}_{pr}$, where k = 15N -s/m. Calculate the minimum speed u_{min} of the block on platform A so that plank reaches the platform B. (Answer in m /s)



7. When a resistance R is connected in series with an element A, the electric current is found to be lagging behind the applied AC voltage V by an angle 30º.When the same resistance is connected in series with element B, current leads by 60º.When R, A, B are connected in series, the current now leads voltage by θ which is equal to $\tan^{-1}(K/\sqrt{3})$, then the value of K is (assume same AC source is used in all cases)

8. The wedge shown in the figure can slide without friction on a horizontal tabletop. The mass of the wedge is M and its angle of elevation is $\alpha = 30^{\circ}$. A body of mass m slides down without friction along the wedge, the path of the body makes an angle of $\phi = 60^{\circ}$ with the ground. Find the ratio of the masses, m/M. (Initially system is at rest)



PART-2 : CHEMISTRY

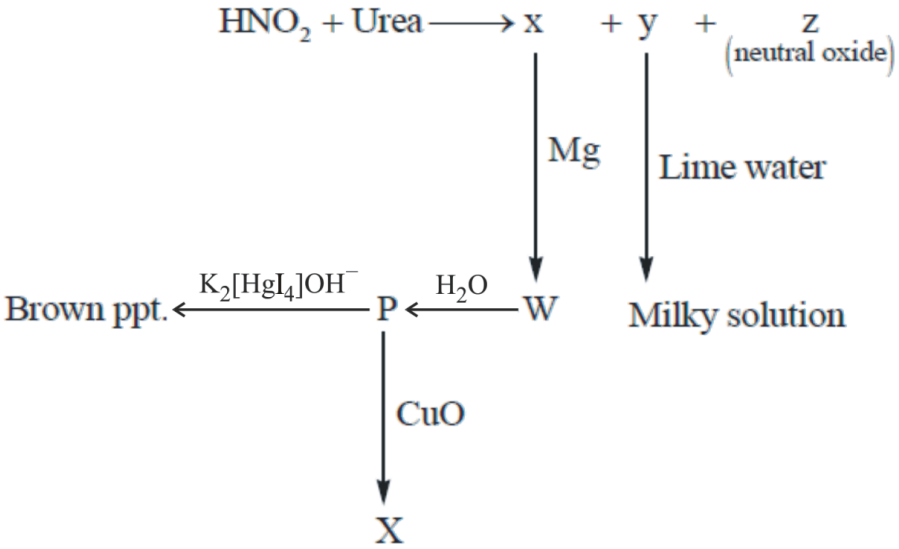
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

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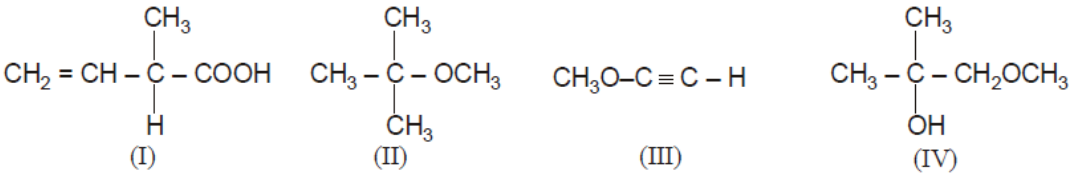
1. Analyse the following reaction sequence.



Which of the following compound(s) on reaction with Cl_2 can produce a compound X, That is used in quantitative estimation of "Magnesium" using carius method?

(A) X (B) W (C) Z (D) P

2. Consider the following compounds and decide as to which of the following statements are true?



(A) (II) gives no reaction with Na metal, however, 1 mole of (IV) on reaction with Na metal will liberate 22.4 litres of H_2 gas at 1atm & 273K

(B) (I) will give brisk effervescence on addition of $NaHCO_3$ but will not bring any change in the colour of Br_2 water

(C) (III) liberates H_2 gas with Na metal, gives white precipitate with Tollen's reagent but does not respond towards Lucas reagent or 2, 4-DNP test.

(D) (IV) gives turbidity with anhydrous $ZnCl_2$ + HCl

3. A 0.1 M solution of Cu^{+} ions is stirred with an excess of potassium cyanide sufficient to convert all the Cu^{+} ions to the corresponding cuprocyanide complex $[Cu(CN)_4]^{-3}$ and in addition to providing the solution with an excess of CN^{-} equal to 0.2 M. Calculate the maximum pH of the solution when the final solution is treated with hydrogen sulphide to maintain $[H_2S] = 0.1M$ and such that the precipitation of cuprous sulphide is prevented. The instability constant for $[Cu(CN)_4]^{-3}$ is $6.4 \times 10^{-15}\ M^4$, $K_{a,\ overall}$ of $H_2S = 1.6 \times 10^{-21}$ M, K_{sp} of $Cu_2S = 2.56 \times 10^{-27} M^3$.

(A) 4.0

(B) 10.0

(C) 10.8

(D) 3.2

4. n containers having volumes v, 2v, 3v.....nv containing gaseous moles n, 4n, 9n,.....n^3 respectively are connected with stopcock at same temperature. If pressure of first container is P, then the final pressure when all the stopcocks are opened is

(A) $\frac{n(2n+1)P}{3}$

(B) $\frac{(2n+1)P}{3n}$

(C) $\frac{(2n+1)P}{6n}$

(D) $\frac{(2n+1)P}{3}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. Which of the following statements is/are correct regarding solubility in water ?

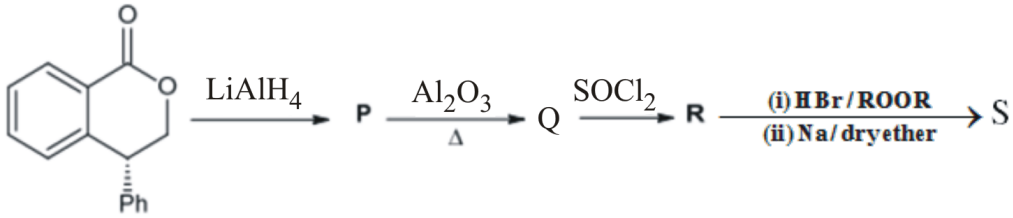
(A) Hexan-1-ol is less soluble than cyclohexanol

(B) 4-methyl phenol is more soluble than heptan-1-ol

(C) 3-Ethyl hexanol is more soluble than octan-2-ol

(D) Cyclooctane-1, 4-diol is more soluble than hexan-2-ol

6. With reference to the scheme given below, which of the following statement(s) is/are true about compounds (A), (B), (C) and (D)?



(A) P is chiral and is optically pure.

(B) Compound Q gives a resolvable mixture on reaction with $Hg(OAc)_2$ followed by reduction with $NaBH_4$

(C) Compound R exhibits geometrical isomerism.

(D) S is a mixture of diastereomers.

7. Which of the following statement(s) is (are) **incorrect**?

(A) A green coloured solution is obtained when thiourea is tested for nitrogen in lassaigne test.

(B) It is not necessary to carry out the chromyl chloride test in a dry test tube.

(C) In $Pb(NO_3)_2$, the brown ring test can be performed with its water extract

(D) Suspension of $CdCO_3$ gives black ppt. with sodium sulphide solution.

8. Which of the following statement(s) is/are **true**?

(A) Soluble bicarbonates give white precipitate with $MgCl_2$ in cold.

(B) Soluble calcium bicarbonates give white precipitate with dilute ammonium solution.

(C) All bicarbonates are generally soluble in water.

(D) Hg(II) chloride forms a reddish brown precipitate in a solution of sodium carbonate.

9. A 150 mL solution of I_2 is divided into two unequal parts. 'I' part reacts with hypo solution in acidic medium where 15 mL of 0.4 M hypo was consumed. 'II' part was added to 100 mL of 0.3 M hot and concentrated NaOH solution. Residual base required 10 mL of 0.3 M H_2SO_4 solution for complete neutralization. Then

(A) Number of moles of I_2 in part-I is 3×10^{-3}

(B) Initial concentration of I_2 is 0.1M

(C) Number of moles of I_2 in part-II is 15×10^{-3}

(D) Number of moles of I_2 in part-II is 12×10^{-3}

10. In a large sample of Hydrogen atoms, all atoms are in 9th excited state. Identify the correct statements when all the excited atoms reach ground state without producing any spectral line in Balmer series.

(A) Total 36 different spectral lines are produced

(B) Total 37 different spectral lines are produced

(C) 8 different spectral lines are produced in Lyman series

(D) 9 different spectral lines are produced in Lyman series

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

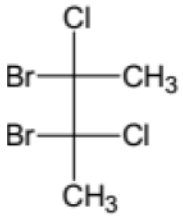
SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If only the correct answer is given.
Zero Marks	:	0	If no answer is given.
Negative Marks	:	–1	In all other cases

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1. The total number(s) of stable conformers with non-zero dipole moment for the following compound is/are

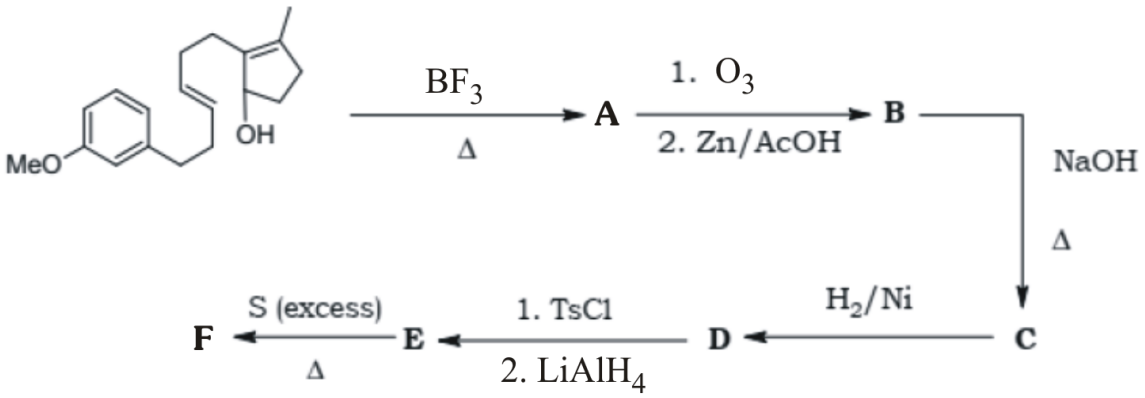


2. The total number of compounds having $p\pi-d\pi$ bonding among the molecules given below are
SO_2, SO_3, CO_2, H_3PO_4, H_3BO_3, $HClO_4$, HNO_3, HOCl

3. A solution of Hg^{2+} ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline precipitate. The coordination number of mercury in the deep blue coloured compound is

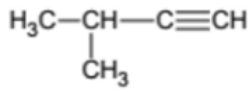
4. $Cl_2 + F_2$(excess) $\xrightarrow{(573K)}$ A $\xrightarrow{\text{Uranium}}$ B + C inter halogen compound
The total no. of ℓ.p's in A is $\longrightarrow$ x
The total no. of ℓ.p's in B is $\longrightarrow$ y
The total no. of ℓ.p's in C is $\longrightarrow$ z
x + y – z = 15 + b. Then 'b' is _____

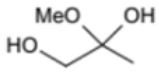
5. Consider the following reaction sequence.

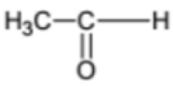


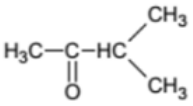
The degree of unsaturation of compound E is

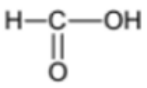
6. How many of the following compounds exhibit positive "Silver mirror test" with Tollen's reagent?

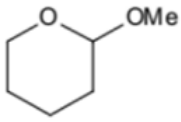
(a) 

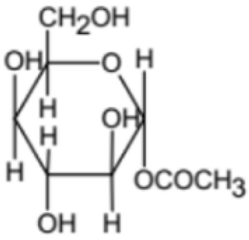
(b) 

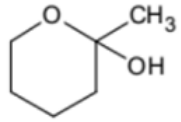
(c) 

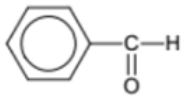
(d) 

(e) 

(f) 

(g) 

(h) 

(j) 

(i) 3, 3−dimethyl-1-butyne

(k) Acetophenone

7. From the following choose the number of correct statement(s) :-

(i) Stability of lyophillic colloids is mainly due to the strong interaction between dispersed phase and dispersion medium.

(ii) Entropy change for micelle formation in aqueous solution is positive.

(iii) Gelatin has considerably low value of gold number and is an effective protective agent.

(iv) Zeta potential is also responsible for stability of lyophobic colloids.

(v) Surface tension of lyophilic colloids is lesser than that of the dispersion medium.

(vi) A positively charged ion can cause the precipitation of As_2S_3 sol

(vii) Cellulose solution is an example of multimolecular colloid system.

(viii) The combination of two layers of same charge around the colloidal particle is called Helmholtz electrical double layer.

8. Number of crystal systems having at least 2 interfacial angles equal = x

Number of crystal systems having none of the interfacial angles equal = y

Number of crystal systems having none of the edge lengths equal = z

Then, x − y + z is_____

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : –1 In all other cases

---

1. If the curves $y = 2(x - a)^2$ and $y = e^{2x}$ touches each other, then 'a' is less than-

(A) -3 (B) -2 (C) -1 (D) 0

2. If normal to the curve $y = x^2$ forms the shortest chord, then the square of slope of normal is equal to

(A) $\frac{1}{\sqrt{2}}$

(B) $\frac{1}{2}$

(C) $\sqrt{2}$

(D) 2

3. If $I(k) = \int_0^{\infty} \frac{\ell nx}{x^2 + kx + k^2} dx$ $(k > 0)$ then

(A) $I(k) > \frac{\ell nk}{3k}$

(B) $I(k) < \frac{\ell nk}{4k}$

(C) $I(1) = 1$

(D) $I(1) < \frac{\pi}{\sqrt{3}}$

4. Let z and w be two complex numbers such that $|z| \leq 1$, $|w| \leq 1$ and $|z + \overline{iw}| = |z - iw| = 2$ then which of the following is **incorrect**?

(A) Number of complex number z satisfying above condition are 2

(B) $Re(z) = Im(w)$

(C) $Im(z) = Re(w)$

(D) $|Z| = |w| \neq 1$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. If $\lim\limits_{n\to\infty}\left(\frac{1}{1^2}+\frac{1}{2^2}+\ldots+\frac{1}{n^2}\right)=\frac{\pi^2}{6}$ and $S=\lim\limits_{n\to\infty}\left(\frac{1}{1^3.2^3}+\frac{1}{2^3.3^3}+\frac{1}{3^3.4^3}+\ldots+\frac{1}{n^3(n+1)^3}\right)$ then $S+\pi^2$ is.

(A) A rational number

(B) An irrational number

(C) An integer

(D) Non integer

6. $f : N \longrightarrow A$ and $g : N \longrightarrow B$ are two onto functions such that $f(n) = \left[n\sec^2\theta\right]$ and $g(n) = \left[n\,\mathrm{cosec}^2\theta\right]$, for some θ. If $\sec^2\theta$ is irrational, then (where[.] denotes the greatest integer function & N is set of natural numbers)

(A) $A \cap B = \phi$

(B) $A \cup B = N$

(C) f and g are bijective

(D) $A \cap B$ is non-empty finite set

7. 16 players $P_1, P_2, \ldots\ldots P_{16}$ play a knock out tournament. Assuming that the players are paired at random in each round. If all the players are of equal strength, then the probability that P_1 reaches final under the condition P_2 reaches semifinal but not in final is $\frac{a}{b}$ (where a and b are co-primes), then which of the following is true?

(A) $\frac{a}{b} = \frac{1}{15}$

(B) $\frac{a}{b} = \frac{2}{15}$

(C) Number of divisors of $a^2b = 12$

(D) Sum of all devisors of $a^2b = 24$

8. Given $\vec{A}(x) = x\hat{i} + x\hat{j} + 5\hat{k}$, $\vec{B}(x) = \hat{i} + x\hat{j}$, $\vec{C}(x) = \hat{i} + \frac{1}{5}\hat{j} + x\hat{k}$. Let α, β, γ denotes three values of x for which $\vec{A}(x)$, $\vec{B}(x)$ and $\vec{C}(x)$ are coplanar, then which of the following is correct?

(A) $[\alpha] + [\beta] + [\gamma] = -1$

(B) $[\alpha] + [\beta] + [\gamma] = 0$

(C) Global maximum value of volume of parallelpiped formed by three vectors whose conterminous edges are $\vec{A}(x)$, $\vec{B}(x)$ & $\vec{C}(x)$ is 4 for $x \in [-2, 3]$

(D) $(1 + \alpha)(1 + \beta)(1 + \gamma) = -4$ (where [.] denotes greatest integer function)

9. A triangle with vertices P(2, 5), Q(5, 2) and R(−1, −1) is inscribed in the rectangular hyperbola $xy - x - y - 3 = 0$ and let $H \equiv (\lambda, \lambda)$ is orthocenter of ΔPQR, let A, B and C be the feet of perpendicular from P, Q and R on QR, PR and PQ respectively and r is in radius of ΔABC, then

(A) Circumcenre of ΔPQR is $\left(-\frac{3}{2}, -\frac{3}{2}\right)$

(B) Incentre of ΔABC is (3, 3)

(C) Equation of AB is $x + y - \frac{26}{5} = 0$

(D) $r = \frac{2\sqrt{2}}{5}$

10. Let $f(x) = \ell n(2x - x^2) + \sin\frac{\pi x}{2}$, then

(A) Graph of f is symmetric about the line x = 1

(B) Graph of f is symmetric about the line x = 2

(C) Local maximum value of f is 1

(D) Local minimum value of f does not exist

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks	:	+3	If only the correct answer is given.
Zero Marks	:	0	If no answer is given.
Negative Marks	:	–1	In all other cases

---

1. Find sum of all integral values of r such that all roots of the polynomial $f(x) = x^3 - (r-1)x^2 - 11x + 4r$ are also integer

2. Let $I = \int_0^{\frac{\pi}{2}} \frac{\cos x + 4}{3\sin x + 4\cos x + 25}dx$ and $J = \int_0^{\frac{\pi}{2}} \frac{\sin x + 3}{3\sin x + 4\cos x + 25}dx$. If $(11I + 2J) = a\pi + b\ln\frac{c}{d}$ where a, b, c and $d \in N$ and $\frac{c}{d}$ is not a perfect square of a rational then find the value of $(a + b + c + d - 55)$.
(Where N is set of natural numbers)

3. Let a vector $\bar{V}_1 = \overline{OP}$, where O is origin and P is (2, 0,2). Vector $\bar{V}_1$ is rotated about origin by an angle of $\frac{\pi}{3}$ in a plane which is perpendicular to XOZ plane to get vector $\bar{V}_2$ where $\bar{V}_2 . \hat{j} > 0$. Again $\bar{V}_1$ is rotated about origin by right angle in XOZ plane to get vector $\bar{V}_3$. If V is volume of tetrahedron whose coterminous edges are $\bar{V}_1$, $\bar{V}_2$ and $\bar{V}_3$ then value of $\frac{3V}{\sqrt{6}}$ is.

4. Suppose that f (x) is a differentiable invertible function with $f(x) \neq 0$ and $h(x) = \int_1^x f(t)dt$.Given that $f(1) = f'(1) = 1$ and g(x) is inverse of f(x). Let $G(x) = x^2 g(x) - xh(g(x)) \forall x \in R$ then $G''(1) =$

5. Let $y = f(x)$ $(f : R \longrightarrow R)$ be an explicit function defined by the implicit equation
$x^3 + y^3 + 3(x^2 + y^2) + 3(x + y) = 14$ and g be the inverse function of f.
If $\frac{d}{dx}[f(x + g(x)).\ g(x + f(x))]$ at $x = -1 + \sqrt[3]{15}$ is equal $\lambda(15)^{\frac{2}{3}}$, where $\lambda \in I$, then find the value of $|\lambda|$.

6. If < a, b, c > are direction cosines of line of intersection of three planes $x + y - z = 1$, $(m-1)x + (3m-7)y - 5z = -3$ and $(m-2)x + (2m-5)y = -4$ for some $m \in R$, then value of $\frac{a+b}{c}$ is _____.

7. Consider a complex number $z = \frac{\sqrt{5}+1}{4} + i\left(\frac{\sqrt{5}-1}{2}\right)\cos\frac{\pi}{10}$ and a set $S = \{Z_p | Z_p = Z^p, p = 0, 1, 2, 3, \ldots .9\}$ Number of ordered pairs (z_i, z_j) where $z_i, z_j \in S$ such that $\arg(z_i) - \arg(z_j) = \frac{2\pi}{5}$ (where $\arg z \in [0, 2\pi)$ is.

8. A curve $y = f(x)$ is passing through (2, 0) and slope of its tangent at any point (x, y) is $\frac{(x+1)^2 + y - 3}{x+1}$. Then the area bounded by the curve $y = f(x)$ and $x -$ axis is $\frac{m}{n}$ where H.C. F of m & n is 1. Then value of $m + n$ is.

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them.	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.
_____	_____
Signature of the Candidate	Signature of the Invigilator

Space for Rough Work



ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)
FULL SYLLABUS
01-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY — PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,B,D	C,D	A,B	A,B,C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	D	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.16 to 0.19	0.11 to 0.13	0.40	0.50	2.40	6.05 to 6.06	4.40 to 4.60	0.27 to 0.29

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A,B,C,D	A,C	A,B	A,B,C,D	D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	3.00	10.00	138.00	4.00	1.00	348.16	40.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C	A,C,D	A,B	A,B,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	B	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.00	13.00	7.00	1.00	8.00	3.00	8.00	4.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,C,D)

If they collide, their vertical component of velocities should be same,

i.e., $100\sin\theta = 160\sin 30^\circ$

$\Rightarrow \sin\theta = 4/5$

Their vertical components will always be same.

Horizontal components.

And $160\cos 30 = 80\sqrt{3}\ ms^{-1}$

They are not same, hence their velocities will not be same at any time. So (b) is correct.

$x = x_1 - x_2 = 160\cos 30^\circ t - 100\cos\theta t$

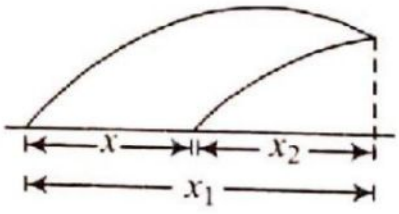
Time of flight: $T = \dfrac{2\times 160\times \sin 30}{g}$

= 16s(same for both)

Now $t < T_x \longrightarrow$ to collide in air

$\Rightarrow \dfrac{x}{80\sqrt{3}-60} < 16 \Rightarrow x < 1280\sqrt{3} - 960$

Since their times of flight are same, they will simultaneously reach their maximum height. So it is possible to collide at the highest point for certain values of x.



2. Ans (A,B,D)

According to Pascal's law, $P_D = P_B$

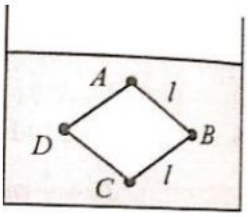
Because B and D lie on the same horizontal plane.

Let ℓ be the side of the square. Then the vertical distance between A and C is $\dfrac{\ell}{\sqrt{2}}$

$P_B = P_V = P_C - \rho g\dfrac{L}{\sqrt{2}}$ ..... (i)

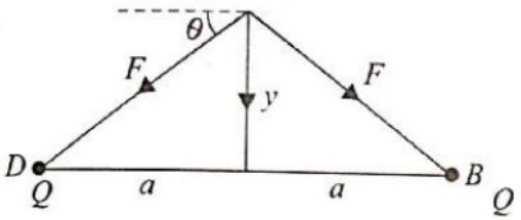
Similarly, $P_B = P_D - P_A + \rho g\dfrac{L}{\sqrt{2}}$ ..... (ii)

$P_B = P_D = \dfrac{P_A + P_C}{2}$



3. Ans (C,D)

The particle will oscillate or perform SHM if equilibrium is a stable one. For negative charge, equilibrium is stable, if the particles are displaced along CD and unstable for displacement along AB.



For y << a, in displaced position, net resultant force toward equilibrium position is

$F_{res} = 2F\sin\theta = \dfrac{2\times Q}{4\pi\varepsilon_0(a^2+y^2)} \times \dfrac{y}{(a^2+y^2)^{\frac{1}{2}}}$

$= \dfrac{2Qy}{4\pi\varepsilon_0 a^3}[a >> y]$

$F_{\sqrt{es}} \propto y$, so SHM for y<<a

For y = a, there would be restoring force, but F is not proportional to y. So there is oscillating motion but not SHM.

4. Ans (A,B)

Let d = distance of the target T from the point of projection. P will strike T if d is an integral multiple of the pitch.

$$\text{Pitch} = \left(\frac{2\pi m}{QB}\right) v\cos\theta = k\left(\frac{v}{B}\right)$$

Where k = constant

Initially, pitch $d = k\left(\frac{v_0}{B_0}\right)$

For the given options if pitch $d' = \frac{d}{2}, \frac{d}{3}, \frac{d}{4}, \ldots\ldots$ etc, then change will hit the target

5. Ans (A,B,C,D)

Δx at O = d [path difference is maximum at O]

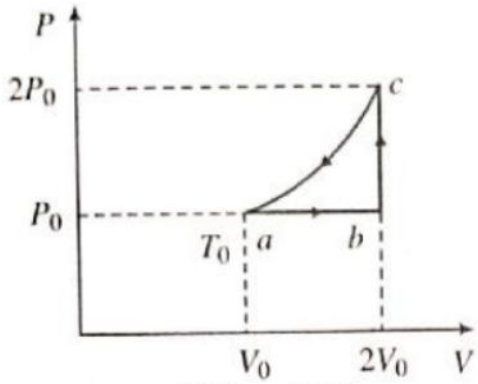
So, if $d = \frac{7\lambda}{2}$, O will be a minima and hence intensity is minimum.

If $d = 4.8\lambda$, then total 10 minima can be observed on the screen, 5 above O and 5 below O, which correspond to

$$\Delta x = \pm\frac{\lambda}{2}, \pm\frac{3\lambda}{2}, \pm\frac{5\lambda}{2}, \pm\frac{7\lambda}{2}, \pm\frac{9\lambda}{2}$$

6. Ans (A,B,C,D)

One mole of an ideal monoatomic gas (initial temperature T_0) is made to go through the cycle abca shown in figure. U denotes the internal energy.



For the process ab, $\frac{P_0V_0}{T_0} = \frac{2P_0V_0}{T_0}$

$T_b = 2T_0$

$T_b > T_a \Rightarrow U_b > U_a$

$$U_b - U_a = C_V\Delta T = \frac{3R}{2}(2T_0 - T_0) = \frac{3RT_0}{2}$$

For the process bc,

$$\frac{P_0(2V_0)}{2T_0} = \frac{2P_0(2V_0)}{T_c} \Rightarrow T_c = 4T_0$$

$T_C > T_0$

$$U_c - U_b = \frac{3R}{2}(4T_0 - T_0) = \frac{9RT_0}{2}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

Let the acceleration of various blocks are as shown in Figure. Pulley P_2 will have downward acceleration a.

Now $a = \frac{a_1 + a_2}{2} \Rightarrow a_2 = 2a - a_1 > 0$

So acceleration of 2 is upwards

Hence, (i) $\longrightarrow$ (b, c)

And $a = \frac{-a_1 + a_4}{2} \Rightarrow a_4 = 2a + a_1 > 0$

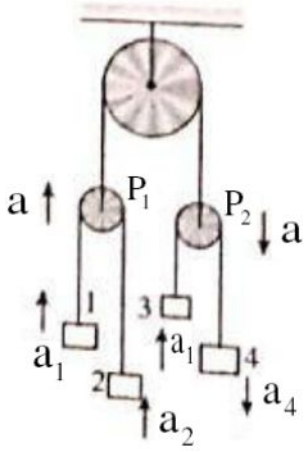
So, acceleration of 4 is downwards.

Hence, (ii) $\longrightarrow$ (a, d)

Acceleration of 2 w.r.t 4

$a_{\frac{2}{4}} = a_2 - (-a_4) = 4a > 0$

This is upwards, Hence, (iv) $\longrightarrow$ (c)



9. Ans (D)

(P) $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$

$= 10 : 80 : 10 = 0.5 : 4 : 0.5$

$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$

$V = \sqrt{\frac{Y}{\rho}} = \frac{5000}{0.4} = 12500 = 12.5\,\text{kHz}$

(Q) $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$

$= 30 : 40 : 30 = 1.5 : 2 : 1.5$

$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$

$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6 \times 10^{11}}{2500}} = 8000\text{m/s}$

$f = \frac{v}{\lambda} = \frac{8000}{0.4} = 20000 = 20\ \text{kHz}$

(R) $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$

$= 10 : 60 : 30 = 0.5 : 3 : 1.5$

$\Rightarrow \frac{5\lambda}{2} = 1 \Rightarrow \lambda = 0.4\text{m}$

$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2 \times 10^{11}}{8000}} = 5000\text{m/s}$

$f = \frac{v}{\lambda} = \frac{5000}{0.4} = 12500 = 12.5\ \text{kHz}$

(S) $n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$

$= 15 : 60 : 25 = 1.5 : 6 : 2.5$

$\Rightarrow \frac{10\lambda}{2} = 1 \Rightarrow \ell = 0.2\text{m}$

$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{1.6 \times 10^{11}}{2500}} = 8000\,\text{m/s}$

$f = \frac{v}{\lambda} = \frac{8000}{0.2} = 40000 = 40\ \text{kHz}$

10. Ans (D)

Maximum kinetic energy of ejected electron is given by Einstein's photoelectric equation $K_{max} = hf - \left(\frac{hf}{3}\right) = \frac{2hf}{3}$. As no potential difference is applied across target and collector and vaccum is there in the tube, so this maximum KE remains same at all locations. b, c and d. Kinetic energy of ejected photoelectrons can be anything from 0 to K_{max} (as found for a). It remains the same at all locations (reasoning is same as for above).

PART-1 : PHYSICS

SECTION-II

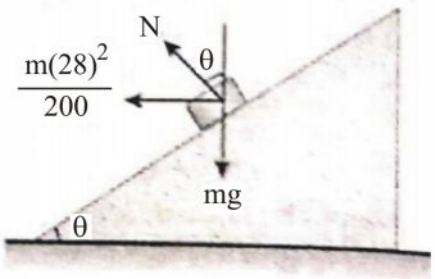
1. Ans (0.16 to 0.19)

When there is not tendency to slip, no frictional force acts. Hence

$N \sin\theta = \frac{m(28)^2}{200}$ 1

and $N \cos\theta = mg$ 2

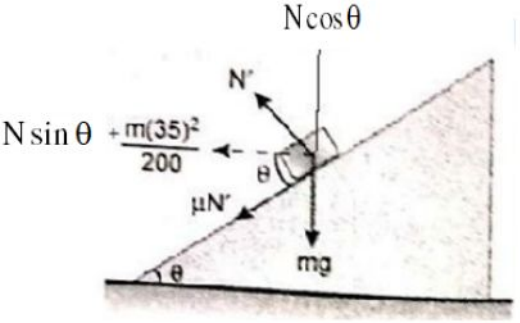
Dividing (1) by (2), we get $\tan\theta = 0.4$ 3



At speed 35 m/s, car has a tendency to slip up the track. Hence frictional force acts down the plane.

Therefore, $N' \sin\theta + \mu N' \cos\theta = \frac{m(35)^2}{200}$ 4

and $N'\cos\theta - \mu N'\sin\theta = mg$ 5



Solving (3), (4) and (5), we get $\mu = 0.18$

2. Ans (0.11 to 0.13)

Initial kinetic energy $\frac{1}{2} \times 2 \times 10^2 = 100J$

Final KE = 50 J

$\therefore \frac{1}{2}mv^2 = 50J$

When the spring is being compressed, the equation of motion of spring mass system can be written as

x Asinωt where $\omega = \sqrt{\frac{k}{m}}$

$\therefore \frac{dx}{dt} = A\omega\cos\omega t$

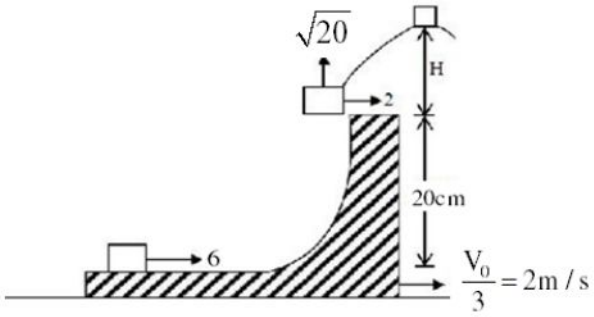
$\Rightarrow v = v_0 \cos\omega t$

$\Rightarrow 5\sqrt{2} = 10\cos\omega t$

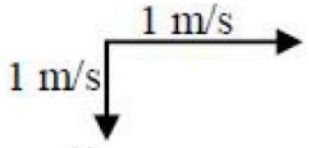
$\Rightarrow \omega t = \frac{\pi}{4}$

$\Rightarrow t = \frac{\pi}{4\omega} = \frac{\pi}{4}\sqrt{\frac{m}{k}} = \frac{\pi}{4}\sqrt{\frac{2}{80}} = 0.124\,\text{sec}$

3. Ans (0.40)



At highest point combined mass velocity is



$H = \frac{20}{2g} = 1m$

$1.2 = 1.T + \frac{1}{2}gT^2$

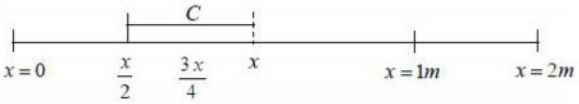
T = 0.4 sec

$u_{rel} = 1$ m/s

$\Rightarrow \ell = u_{rel}\, T = 0.4 = 40cm$

4. Ans (0.50)

The co-ordinates of centre of mass (point C) of the moving part is $\frac{3x}{4}$



$\therefore\ X_c = \frac{3x}{4}$

$\therefore\ v_c = \frac{d}{dt}(X_c) = \frac{3}{4}\frac{dx}{dt} = \frac{3}{4}\text{m/s}\left(\text{as } \frac{dx}{dt} = 1\text{m/s}\right)$

Linear momentum of moving point is P = mv

Where v =1m / s and m is increasing uniformly with time.

The net force on the moving part is thus,

$F = \frac{dP}{dt} = \frac{dm}{dt}v + \frac{dv}{dt}m$

or $F = \left(\frac{dm}{dt}\right)(1) + 0$

Here $\frac{dm}{dt}$ can be found with the help of following argument.

The moving end of the carpet starts from the origin and the whole carpet will be moving when it reaches x = 2m.This will happen after 2 seconds. Hence $\frac{dm}{dt} = \frac{1}{2}\text{kg/s}$

$\therefore\ F_{min}$ (neglecting all dissipative forces) $= \frac{1}{2}$ N

5. Ans (2.40)

Bulk modulus is given by $B = -\frac{dP}{(dV/V)}$ or $\frac{dV}{V} = \frac{dP}{B}$

or $P.\frac{dV}{V} = -\frac{PdP}{B}$

$\frac{PdV}{V}$ is the work done per unit volume. The negative sign implies that a decrease in pressure gives rise to increase in volume and vice-versa.

Hence volume density of elastic potential energy $u = \int_0^{h\rho g}\frac{P.dP}{B} = \frac{1}{2}\frac{(h\rho g)^2}{B}$

Substitution the value, we obtain

$u = \frac{1}{2}\frac{(10^3)(10^3)^2.(9.8)^2}{2\times 10^9}$

$\Rightarrow u = 2.4\times 10^4\ \text{J/m}^3$

6. Ans (6.05 to 6.06)

Applying Snell's law on face AB,

$1\sin 45° = \sqrt{2}\sin r$

$\therefore \sin r = \frac{1}{2}$

or r = 30°

i.e., may becomes parallel to AD inside the block.

Now applying, $\frac{\mu_0}{v} = \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ on face CD $= \frac{1.514}{OE} - \frac{\sqrt{2}}{\infty} = \frac{1.514 - 1.414}{0.4}$

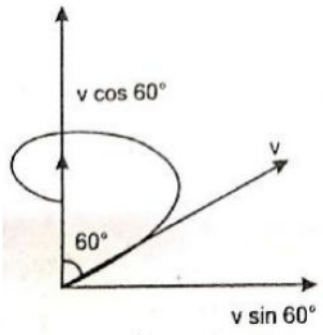
Solving this equation, we get OE = 6.06m

7. **Ans (4.40 to 4.60)**

Pitch of the helix is given by

$$y = (v\cos\theta).T = v(\cos 60^\circ)T = \frac{vT}{2}$$

Where time period of circular motion $T = \frac{2\pi m}{eB}$



$$\text{Pitch } y = \left(\frac{2\pi m}{Be}\right) v\cos 60^o = \left(\frac{\pi mv}{Be}\right)$$

$$\therefore \text{ KE} = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{Bey}{\pi m}\right)^2$$

Using Einstein's equation

$$\frac{1}{2}mv^2 = hv - \phi$$

$$\phi = hv - \frac{1}{2}m\left(\frac{Bey}{\pi m}\right)^2$$

$$= \left[4.9 - \frac{1}{2}\left(\frac{m}{e}\right)\left(\frac{Bey}{\pi m}\right)^2\right]$$

electron – volts = 4.9 – 0.4 = 4.5eV

8. **Ans (0.27 to 0.29)**

The product nucleus ^{198}Hg is in excited state and possesses extra 1.088 MeV energy.

If ^{198}Hg would had been in ground state, the kinetic energy available to electron and antineutrino must have

$Q = (m_{Au} - m_{Hg})\,931\text{MeV}$

$= (197.968233 - 197.966760)931 \text{ MeV}$

$= 1.3714 \text{ MeV}$

Since ^{198}Hg is in excited state, actual kinetic energy available to electron and antineutrino is

$K = (1.3714 - 1.088)\text{MeV}$

$= 0.283 \text{ MeV}$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A,B,C)**

Metals of 2nd & 3rd transition series form low spin complexes with weak or strongfield ligand and form square planar complexes with complex number 4.

Ni(II) with strong ligand like CN^- form square planar complexes.

Hg(II) form tetrahedral complexes since it has d^{10} config.

2. **Ans (A,B,C,D)**

Froth floatation method is a physical method depending on adsorption of potassium ethyl xanthate which prevents the wettability of or particle. It is used only for the concentration of sulphide ores where ore particles come into frothwhile impurities sink down.

3. **Ans (A,C)**

Both S_N1 and S_N2 are not possible on vinyl halides and aryl halides.

4. **Ans (A,B)**

Value based.

5. **Ans (A,B,C,D)**

Variation of composition changes internal energy of the system.

6. **Ans (D)**

$AgI_{(s)} \rightleftharpoons Ag^+_{(aq)} + I^-_{aq}$ $\qquad \Delta^oG_1 = -RT\ln K_{sp}$

$Ag^+_{(aq)} + e^- \rightleftharpoons Ag_{(s)}$ $\qquad \Delta^oG_2 = -nF\,E^o_{Ag^+|Ag}$

$AgI_{(s)} + e^- \rightleftharpoons Ag_{(s)} + I^-_{(aq)}$ $\qquad \Delta^oG_3 = -nF\,E^o_{I^-|AgI|Ag}$

$\Delta^oG_3 = \Delta^oG_1 + \Delta^oG_2$

Substituting, we get

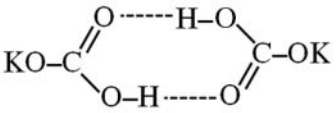
$$\log K_{sp} = \frac{-0.952}{0.059} = -16.13$$

PART-2 : CHEMISTRY

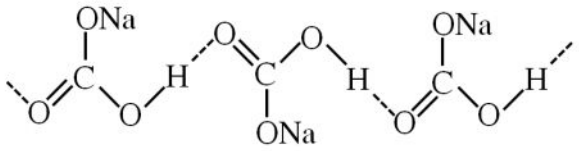
SECTION-I (ii)

7. Ans (A)

$KHCO_3$ exist as dimer.



$NaHCO_3$ exist in polymeric form.



$LiHCO_3$ exist only solution and exist as Li^+ and HCO_3^- similar to $Mg(HCO_3)_2$

(diagonal relationship).

8. Ans (D)

(A) $H_2P_2O_6^{2-}$ have 2 ionisable hydrogen. The phosphorous atoms are involved in sp^3 hybridization. It is non–planar species.

(B) Similarly $H_2P_2O_7^{2-}$ also have 2 ionisable hydrogens, sp^3 hybrid p atoms and non–planar structure.

(C) $H_2S_2O_5$ have 2 ionisable hydrogens, sp^3 hybrid sulphur atoms with non–planar structure.

(D) HSO_5^- ion has only one ionizable hydrogen in which S–atom is involved in sp^3 hybridization with non–planar structure. It can act as oxidizing agent.

9. Ans (A)

Direct reactions.

10. Ans (B)

All are fcc lattices. Symmetry is followed.

PART-2 : CHEMISTRY

SECTION-II

1. Ans (1.00)

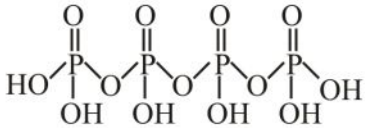
NaOH, KOH, $Ba(OH)_2$, $Sr(OH)_2$ and TlOH are strong bases. $Be(OH)_2$, $Al(OH)_3$, $Sn(OH)_2$ and $Zn(OH)_2$ are amphoteric and these react with each other forming salts that are soluble.

$Mg(OH)_2$ is insoluble in bases. So only $Mg(OH)_2$ remain in the residue. One base is excess and that also dissolve in water.

2. Ans (3.00)

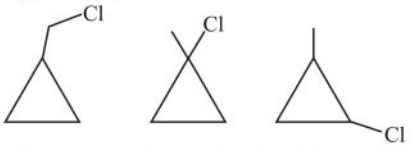
The ratio of the two different types of OH groups should be determined. If one phosphorous atom is attached by two OH groups, their K_{a_1} and K_{a_2} values differ.

The first OH group is a strongly acidic while the second one is weak. If only one OH group is present on one phosphorous atom. It also behave strongly acidic. So phosphorous atoms which contain two OH groups will ionize in two steps. The strongly acidic groups are titrated by the first 30.4. The two terminal OH groups are titrated by the remaining 45.6 – 30.4 = 15.2 CC. This indicates each OH requires 7.6 CC of the titrant. It can be now calculated that there are titrant. It can be now calculated that there are $\frac{30.4}{7.6} = 4$ strongly acidic groups per one polyphosphoric acid. A molecule with two terminal OH groups and four further OH groups is a tetrapolyphosphoric acid.



Thus the number of P – O – P bonds is 3.

3. Ans (10.00)

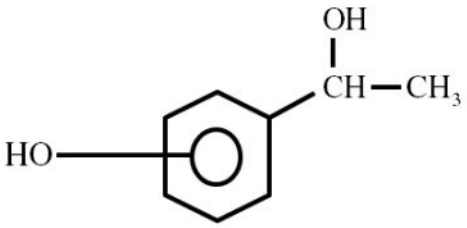


P 1 + 1 + 4 = 6

Q 1 + 1 + 2 = 4

P + Q = 10

4. **Ans (138.00)**



∴ minimum molecular weight = 138

5. **Ans (4.00)**

I, IV, VI, VII

6. **Ans (1.00)**

$$2NO + O_2 \longrightarrow 2NO_2$$

t=0 2 mol 0.5 mol 0

t=t 1.5 mol 0.25 mol 0.5 mol

$$\Delta P = \frac{0.25 \times RT}{V} = \frac{0.25}{12} \times \frac{300}{6.25} = 1 \text{ atm}$$

7. **Ans (348.16)**

$$IE_4(Be) = \frac{Z^2}{n^2} \times 13.6 \times 1.6 \times 10^{-19} J$$

$$= 348.16 \times 10^{-19}$$

8. **Ans (40.00)**

$$\text{Average rate in first 10 min.} = \frac{\Delta p}{\Delta t} = \frac{2}{10} \frac{atm}{min}$$

$$\text{Average rate in next 10 min.} = \frac{\Delta p}{\Delta t} = \frac{2}{10} \frac{atm}{min}$$

∴ Reaction is zero order

$$\therefore t = \frac{8}{K} = \frac{8}{1/5} = 40 \text{ mins}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B,C,D)**

$$\int_1^{xy} f(t)dt = y\int_1^x f(t)dt + x\int_1^y f(t)dt,\ f(1) = 1$$

Differentiating partially w.r.t. 'y'

$$f(xy).x = \int_1^x f(t)dt + xf(y),\ \text{, put } y = 1$$

$$\Rightarrow x\,f(x) = \int_1^x f(t)dt + x$$

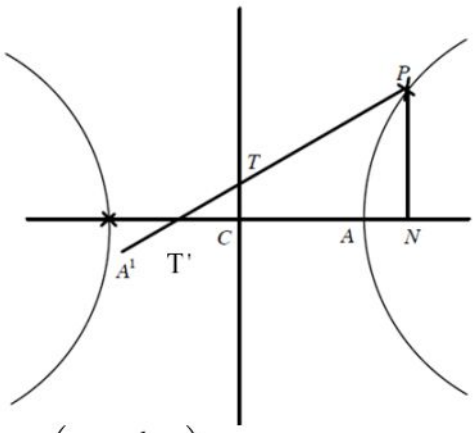
$$\Rightarrow f'(x) = \frac{1}{x}$$

$$\Rightarrow f(x) = \begin{cases} \ln(-x) + c, & x < 0 \\ \ln x + 1, & x > 0 \end{cases}$$

∴ A, B, C, D are true

2. **Ans (A,B,C)**

C=(0, 0) N = (asecθ, 0)



Clearly $T = \left(0, -\frac{b}{\tan\theta}\right)$

$T' = \left(\frac{a}{\sec\theta}, 0\right)$

$CT.PN = b^2$

$CN.CT^1 = a^2$

$CN.NT^1 = AN.A^1N = a^2 \tan^2\theta$

∴ A,B,C are correct

3. Ans (A,C,D)

a, b, c are integers and $a^2 + b^2 + c^2 = 2 \Rightarrow$ one of them is 0 and the other can be ± 1.

$\therefore$ No. of cases = 12

For unique solution $|A| \neq 0 \Rightarrow \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} \neq 0$

$\therefore$ a = 0, b = 1, c = –1 and there are 6 cases

$\therefore$ Probability $= \frac{6}{12} = \frac{1}{2}$

But in all the remaining 6 cases it cannot have infinitely many solutions

For example for a = 0, b =–1, c = 1, $1\ \Delta_1 \neq 0$

$\therefore$ It has no solution

$\therefore$ B is not correct

$a = 0 \Rightarrow b^2 + c^2 = 2$ (1, –1) (–1, 1) it has no solution

$a^2 + b^2 + c^2 - ab - bc - ca \geq 0$

$\Rightarrow ab + bc + ca \leq 2$

$(a + b + c) \geq 0 \Rightarrow ab + bc + ca \geq -1$

But a = b = c not possible

$\therefore$ 2 is not there

$ab + bc + ca \neq 0$

$\therefore$ –1,1 are possible values

4. Ans (A,B)

The plane after rotation is $\begin{vmatrix} x-3 & y-1 & z-2 \\ 2 & 4 & 4 \\ 1 & 4 & 4 \end{vmatrix} = 0$

$\Rightarrow (x - 3)(12) - (y - 1)(6) + (z - 2)(0) = 0$

$\Rightarrow 2x - 6 - y + 1 = 0$

$\Rightarrow 2x - y - 5 = 0$

Equation of plane before rotation

$\Rightarrow \begin{vmatrix} x-3 & y-1 & z-2 \\ 1 & 2 & 2 \\ 2 & -1 & 0 \end{vmatrix} = 0$

$\Rightarrow 2x + 4y - 5z = 0$

5. Ans (A,B,D)

From graph clearly it will have a solution when $a > 0$ and when $a < 0$ then

$\Rightarrow x^2 - a = x$ for atleast one x

$\Rightarrow 1 + 4a \geq 0 \Rightarrow a \geqslant -\frac{1}{4}$

$\therefore$ A, B,D is the key

6. Ans (A,C)

For $\Delta A_1B_1C_1$ the triangle ABC will be the pedal triangle

$\therefore$ Orthocentre of $\Delta A_1B_1C_1$ will be the incentre of ΔABC

$A_1 = 90° - \frac{A}{2} = \frac{\pi}{3} - \left(\frac{A}{2} - \frac{\pi}{6}\right)$

$A_2 = \frac{\pi}{3} - \left(-\frac{A}{2^2} + \frac{\pi}{12}\right)$

$\therefore \lim_{x \to \infty} A_n = \frac{\pi}{3}$

$\therefore \Delta A_nB_nC_n$ becomes equilateral

$B_1C_1 = R \sin\left(90° - \frac{A}{2}\right) = R \cos\frac{A}{2} = \frac{a}{2 \sin A} . \cos\frac{A}{2}$

$= \frac{a}{2.2 \sin\frac{A}{2}\cos\frac{A}{2}} \cos\frac{A}{2} = \frac{a}{4} \operatorname{cosec}\frac{A}{2}$

PART-3 : MATHEMATICS

SECTION-I (ii)

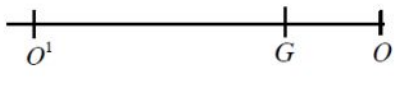
7. Ans (A)

$\overline{SA} + \overline{SB} + \overline{SC} = \overline{OA} + \overline{OB} + \overline{OC} - 3\overline{OS} = 3\overline{OG} - 3\overline{OS} = 3\left(\overline{SG}\right)$

A⟶Q

$\overline{OA} + \overline{OB} + \overline{OC} = 3\overline{OG}$

$3G = 2O + O'$



$|\overline{OO'}| = 3|\overline{OG}|$

$\overline{O'A} + \overline{O'B} + \overline{O'C} = \overline{OA} + \overline{OB} + \overline{OC} - \left(\overline{OO'} + \overline{OO'} + \overline{OC}\right)$

$= 3\overline{OG} - 3\overline{OO'} = \overline{OO'} - 3\overline{OO'} = 2\overline{OO'}$

$\therefore$ Answer is 2.

D) $|\overline{OO'} - \overline{OA} + \overline{OB} - \overline{OO'} + \overline{OC} - \overline{OO'}|$

$|\overline{OA} + \overline{OB} + \overline{OC} - 2\overline{OA} - \overline{OO'}| = |$

$3\overline{OG} - 2\overline{OA} - \overline{OO'} = |2\overline{OA}| = 2R = |\overline{AP}|$

$\therefore$ Answer is 5.

8. Ans (B)

(A) $|\bar{a}\ \bar{b}\ \bar{c}| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}^2$

$= (1(-1)-1(-2)+1(3))^2 = 16$

$A \longrightarrow r$

B) Equation of $\overline{OQR}$ is $\begin{vmatrix} x & y & z \\ 1 & 3 & 4 \\ 2 & 1 & -2 \end{vmatrix} = 0$

$\Rightarrow x(-10) - y(-10) + z(-5) = 0$

$\Rightarrow 2x - 2y + z = 0$

$\therefore$ Distance $= \dfrac{|16+4-11|}{3} = \dfrac{9}{3} = 3$

C) $(\lambda, \lambda-1, \lambda)$, $(2u-1, u, u)$ be A and B

$\therefore$ Dr's of $\overline{AB}$ are $(2u-\lambda-1, u-\lambda+1, u-\lambda)$ $(2, 12)$

$\dfrac{2u-\lambda-1}{2} = \dfrac{u-\lambda+1}{1} = \dfrac{u-\lambda}{2}$

$\Rightarrow \lambda = 3$

$2u - 2\lambda + 2 = u - \lambda$

$u = \lambda - 2 = 1$

A(3, 2, 3) B(1, 1, 1)

$\therefore AB = \sqrt{4+1+4} = 3$

D) Let OABC be the tetrahedron

Centre of $\overline{OAB} = \dfrac{\bar{a}+\bar{b}}{3} = G_1$. Centre of $\overline{OBC} = \dfrac{\bar{b}+\bar{c}}{3} = G_2$

$\therefore G_3 = \dfrac{\bar{c}+\bar{a}}{3}$, $G_4 = \dfrac{\bar{a}+\bar{b}+\bar{c}}{3}$

$\therefore$ Volume of smaller tetrahedron

$= \dfrac{1}{6}\left|\left[\dfrac{\bar{c}}{3}\ \dfrac{\bar{a}}{3}\ \dfrac{\bar{b}}{3}\right]\right|$

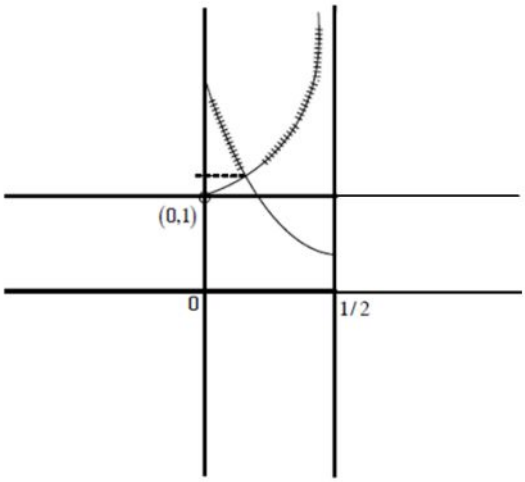
Volume of tetrahedron $= \dfrac{1}{6}|[\bar{a}\ \ \bar{b}\ \ \bar{c}]|$

$\therefore$ Ratio $= \dfrac{1}{27}$

$\therefore m + n = 28$

9. Ans (C)

$0 < x < \dfrac{1}{2} \Rightarrow 0 < \pi x < \dfrac{\pi}{2} \Rightarrow 0 < \sin\pi x < 1$



They intersect at $x = \dfrac{1}{4}$

$\therefore$ Range $= (3-\sqrt{2}, 2] - \{1\}$

Numbers integers = 1

Clearly no.of extreme = 1

At $x = \dfrac{1}{4}$

$f(x)$ is an integer

10. Ans (C)

A) $\sqrt{\sin x}$ is increasing on $\left[0, \frac{\pi}{3}\right]$ so it is one-one

and range = $\left[0, \sqrt{\frac{\sqrt{3}}{2}}\right]$

$\therefore$ into

B) $f(x_1) = f(x_2) \Rightarrow \frac{x_1+3}{x_1-1} = \frac{x_2+3}{x_2-1}$

$\Rightarrow -x_1 + 3x_2 - 3 = 3x_1 - 3 - x_2$

$\Rightarrow 3(x_1 - x_2) + (x_1 - x_2) = 0 \Rightarrow x_1 = x_2$

$y = \frac{x+3}{x-1} \Rightarrow xy - y = x + 3$

$\Rightarrow x = \frac{y+3}{y-1}$

$x > 1 \Rightarrow \frac{y+3}{y-1} - 1 > 0$

$\Rightarrow \frac{4}{y-1} > 0 \Rightarrow y > 1$

$\therefore$ Range = $(1, \infty)$

$\therefore$ One-One, onto

C) sinx is one-one $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\therefore$ It is many-one onto

$f'(x) = \frac{x(x-4)}{(x-2)^2}$

$\therefore$ It is $\uparrow$ on $(4, \infty)$ and $\downarrow$ in $(2, 4)$

$\therefore$ It is many-one

D) $y = \frac{x^2}{x-2} \Rightarrow x^2 - xy + 2y = 0$

$\Rightarrow y^2 - 8y \geqslant 0$

$\Rightarrow$ but $y > 0$

$\therefore y \in [8, \infty)$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$\arg\left(z^{1/3}\right) = \frac{1}{2}\arg\left(z^2 + \bar{z}z^{1/3}\right)$

$\arg\left(\frac{z^{1/3}}{z^2 + \bar{z}z^{1/3}}\right) = 0$

$\Rightarrow \frac{z^{2/3}}{z^2 + \bar{z}z^{1/3}} = \frac{\bar{z}^{2/3}}{\bar{z}^2 + z\bar{z}^{1/3}}$

$\Rightarrow z^{2/3}.\bar{z}^2 + zz^{2/3}z^{-1/3} = z^2\bar{z}^{2/3} + \bar{z}^{5/3}z^{1/3}$

$\Rightarrow z^{2/3}.\bar{z}^{2/3}(\bar{z}^{4/3} - z^{4/3}) + z^{1/3}\bar{z}^{1/3}(z^{4/3} - z^{-4/3}) = 0$

$\Rightarrow (\bar{z}^{4/3} - z^{4/3})(|z|^{2/3} - |z|^{4/3}) = 0$

$\Rightarrow z = \bar{z} \Rightarrow z$ is real not possible

$\therefore |z|^{2/3}(|z|^{2/3} - 1) = 0 \Rightarrow |z| = 1$

2. Ans (13.00)

Equation of circle is $x^2 + y^2 = a^2e^2$,

Let P = $(a\cos\theta, a\sin\theta)$

Area of triangle = 30 $\Rightarrow a^2e^2\sin\theta = 30$ .....(1)

But P lies on the ellipse $\Rightarrow \sin\theta = \frac{1}{e^2} - 1$

$\therefore$ (1) $\Rightarrow \frac{289}{4}e^2\frac{(1-e^2)}{e^2} = 30$

$\Rightarrow e^2 = \frac{169}{289} \Rightarrow e = \frac{13}{17}$

$\therefore S_1S_2 = 2 \times \frac{17}{2} \times \frac{13}{17} = 13$

3. Ans (7.00)

Given

$$\det = \begin{vmatrix} -1 & b & a+2b+2c \\ -1 & c & b+2a+2c \\ -1 & a & c+2a+2b \end{vmatrix} \; c_1 \to c_1 + c_2$$

$$= \begin{vmatrix} -1 & b & a+2b+2c \\ 0 & c-b & a-b \\ 0 & a-c & b-c \end{vmatrix}$$

$= a^2 + b^2 + c^2 - ab - bc - ca$

$= (-1)^2 - 3(-2) = 7$

4. Ans (1.00)

$a_2 = 2(1 + a_1)$

$\Rightarrow 1 + a_1 = \frac{a_2}{2}, 1 + a_2 = \frac{a_3}{3} \ldots\ldots 1 + a_{n-1} = \frac{a_n}{n}$

Given limit =

$$\lim_{n\to\infty}\left(\frac{1+a_1}{a_1}\right)\left(\frac{1+a_2}{a_2}\right)\ldots\ldots\left(\frac{1+a_n}{a_n}\right)$$

$$= \lim_{n\to\infty}\frac{1}{2}.\frac{1}{3}\ldots\ldots\frac{1}{n}(1+a_n) = \lim_{n\to\infty}\frac{(1+a_n)}{n!}$$

But $\frac{a_n}{n!} = \frac{1}{(n-1)!} + \frac{a_{n-1}}{(n-1)!}$

$\Rightarrow \frac{a_n}{n!} - \frac{a_{n-1}}{(n-1)!} = \frac{1}{(n-1)!}$

Put n = 2, 3, 4 ..... $\frac{a_2}{2!} - \frac{a_1}{1!} = \frac{1}{1!}$

$\frac{a_3}{3!} - \frac{a_2}{2!} = \frac{1}{2!}$

$\frac{a_n}{n!} - \frac{a_{n-1}}{(n-1)!} = \frac{1}{(n-1)!}$

$\frac{a_n}{n!} - \frac{a_1}{1!} = \frac{1}{1!} + \frac{1}{2!} + \ldots\ldots + \frac{1}{(n-1)!}$

$\therefore \frac{a_n}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \ldots\ldots + \frac{1}{(n-1)!}$

$\therefore \lim_{n\to\infty}\left(\frac{1+a_n}{n!}\right) = \lim_{n\to\infty}\frac{1}{n!} + \lim_{n\to\infty}\frac{a_n}{n!} = e$

$\therefore$ Required value = $e \times \frac{1}{e} = 1$

5. Ans (8.00)

$f(x) + f(y) \le 0$

$\Rightarrow x^2 + 6x + 1 + y^2 + 6y + 1 \le 0$

$\Rightarrow (x+3)^2 + (y+3)^2 \le 16$

$f(x) - f(y)$

$\Rightarrow x^2 + 6x + 1 - y^2 - 6y - 1 \le 0$

$\Rightarrow (x^2 - y^2) + 6(x - y) \le 0$

$\Rightarrow (x - y)(x + y + 6) \le 0$

Required area is the area of semicircle = 8π

$\therefore k = 8$

6. Ans (3.00)

$\cos^2\frac{\pi}{8} = \frac{\sqrt{2}+1}{2\sqrt{2}}$, the other root is $\frac{\sqrt{2}-1}{2\sqrt{2}}$

$\therefore$ Sum = $1 = -p \Rightarrow p = -1$

$\therefore [\cos 1 + \sin 1] = \left[\sqrt{2}\sin\left(\frac{\pi}{4} + 1\right)\right] = 1$

$q^2 = \frac{1}{8} \Rightarrow 1 - 1 + 2\left|\log_2\frac{1}{2\sqrt{2}}\right| = 3$

7. Ans (8.00)

Let $\bar{a} = (\cos x \cos y, \sin y, \sin x \cos y)$, $\bar{b} = (1, 2, 2)$

$\bar{a}.\bar{b} = \cos x \cos y + 2\sin y + 2\sin x \cos y$

$(\bar{a}.\bar{b})^2 \le |\bar{a}|^2|\bar{b}|^2$

$\Rightarrow (\cos x \cos y + 2\sin y + 2\sin x \cos y)^2 \le 1.9$

$\Rightarrow \cos x \cos y + 2\sin y + 2\sin x \cos y \le 3$

$\therefore \frac{\cos x \cos y}{1} = \frac{\sin y}{2} = \frac{\sin x \cos y}{2}$

$\tan y = 2\cos x$, $\tan y = \sin x$

$\tan x = 2$, $\tan y = \frac{2}{\sqrt{5}}$

$\therefore \tan^2 x + 5\tan^2 y = 4 + 4 = 8$

8. Ans (4.00)

$$\lim_{n\to\infty}\left(\frac{3!}{2n!n!}.\frac{n!n!}{(2n)!}\right)^{\frac{1}{n}} = \frac{a}{b}$$

$$y = \lim_{n\to\infty}\left(\frac{(2n+1)(2n+2)\ldots\ldots(2n+n)}{(n+1)(n+2)\ldots\ldots(n+n)}\right)^{\frac{1}{n}} = \frac{a}{b}$$

$$\lim_{n\to\infty}\left(\frac{\left(2+\frac{1}{n}\right)\left(2+\frac{2}{n}\right)\ldots\ldots\left(2+\frac{n}{n}\right)}{\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\ldots\ldots\left(1+\frac{n}{n}\right)}\right)^{\frac{1}{n}} = \frac{a}{b}$$

$$\log y = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left[\log\left(2+\frac{r}{n}\right) - \log\left[1+\frac{r}{n}\right]\right] = \frac{a}{b}$$

$$= \int_0^1 \log\left(\frac{2+x}{1+x}\right)dx = \int_2^3 \log t\,dt - \int_1^2 \log z\,dz$$

$= (t(\log t - 1))_2^3 - (z(\log z - 1))_1^2$

$=3(\log 3 - 1) - 2(\log 2 - 1) - 2(\log 2 - 1) + 1(0 - 1)$

$=3\log 3 - 4\log 2 = \log\frac{27}{16}$

$\therefore \frac{a}{b} = \frac{27}{16}$

$\frac{a+b+1}{a-b} = \frac{27+16+1}{27-16} = \frac{44}{11} = 4$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)
FULL SYLLABUS
01-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY — PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C	A,B,C	A,D	A,B,C	A,B,C	A,B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	2	8	0	8	2	9	2	2

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	D	C	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C,D	A,B	A,B,C,D	B,C,D	A,B,D	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	4	4	8	7	4	6	8

PART-3 : MATHEMATICS

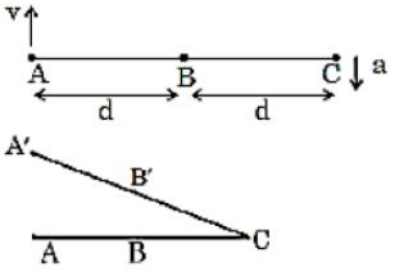
SECTION-I (i)	Q.	1	2	3	4				
	A.	D	B	D	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,B,C	B,C	B,D	B,C,D	A,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	3	4	4	3	0	1	8	7

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)



Let us observe the motion of A and B relative to C.

$$\frac{AA'}{2d} = \frac{BB'}{d}$$

$$BB' = \frac{AA'}{2} = \frac{vt + \frac{1}{2}at^2}{2}$$

$$BB' = \frac{Vt}{2} + \frac{1}{4}at^2$$

$\frac{V}{2}$ is initially speed of B w.r.t C as well as ground.

$\frac{a}{2}$ is acceleration of B w.r.t C

$$\vec{a}_{B/g} = \vec{a}_{B/C} + \vec{a}_{C/g}$$

$$= \frac{a}{2} - a = \frac{-1}{2}a$$

4. Ans (B)

The wavelength at which the intensity of radiation is maximum depends only on the temperature. As this wavelength is the same for both, their temperatures must be equal. The difference in intensities can then be due only to a difference in their emissivities.

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C)

Negative zero error = -1.25

Thickness = $18 + 0.34 - (-1.25) = 19.59$ mm

6. Ans (A,B,C)

$$\text{Speed} = v = \sqrt{\frac{T}{\mu}} = 50\text{m/ s}$$

$$\text{Power} = \frac{1}{2}\mu\omega^2a^2v = 0.126\text{W}$$

$$E_{avg} = P_{avg}.t = P_{avg}\left(\frac{1}{9}\right) = 2.5\text{mJ}$$

7. Ans (A,D)

$$U = \int_R^{2R} \frac{1}{2}\varepsilon_0 E^2 dV = \frac{1}{2}\varepsilon_0 \frac{K^2Q^2}{r^4} 4\pi r^2 dr$$

$$= \frac{KQ^2}{2}\int_R^{2R} r^{-2}dr = \frac{Q^2}{8\pi\varepsilon_0}\left[\frac{1}{-2R} + \frac{1}{R}\right]$$

$$= \frac{Q^2}{16\pi\varepsilon_0 R}$$

Now no electric field inside sphere.

So $U = 0$ (inside)

$$U_{self} = \frac{KQ^2}{2R} = \text{heat lost}$$

8. Ans (A,B,C)

Potential difference is defined only for electrostatic field.

9. Ans (A,B,C)

$$\frac{n_P}{n_P\left(\frac{d}{n_P} + \frac{t}{1} + \frac{x}{n_G}\right)} + \frac{1}{\infty} = \frac{(1 - n_P)}{-R}$$

$$\Rightarrow \frac{d}{n_P} + t = \frac{R}{n_P - 1} - \frac{x}{n_G}$$

10. Ans (A,B,C)

$$Q = \frac{h^2}{24m\lambda^2} = K_B + K_C$$

Where k is the kinetic energy

From conservation of linear momentum

$$\left|\vec{P}_C\right| = \left|\vec{P}_a\right| = P$$

$$\because K = \frac{P^2}{2m}$$

$$\therefore Q = \frac{P^2}{2\times 12m} + \frac{P^2}{2\times 4m}$$

$$\frac{h^2}{24m\lambda^2} = \frac{P^2}{2}\left[\frac{1}{12m} + \frac{1}{4m}\right]$$

$$\frac{h^2}{12m\lambda^2} = P^2\left(\frac{1+3}{12m}\right) = \frac{P^2}{3m}$$

$$\Rightarrow P^2 = \frac{h^2\times 3m}{12m\lambda^2} \Rightarrow P = \frac{h}{2\lambda}$$

$$\Rightarrow \frac{h}{P} = 2\lambda = \text{de-Broglie wavelength}$$

$$\because \left|\vec{P}_B\right| = \left|\vec{P}_C\right|$$

$$\text{Mass defect, } \Delta m = \frac{Q}{C^2} = \frac{h^2}{24mc^2\lambda^2}$$

$$\therefore m_A = 16m + \frac{h^2}{24mc^2\lambda^2}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (2)

$$k = \frac{2\pi}{\lambda} = 62.8\lambda = 0.1m\ell = 40\frac{\lambda}{2} = 2m$$

2. Ans (8)

$Kx\cos 37^\circ = T\cos 53^\circ$

$Kx\cos 53 + T\cos 37 = mg$

$$\Rightarrow \frac{T\left(\cos^2 53^\circ + \cos^2 37\right)}{\cos 37} = mg$$

$$\Rightarrow T = \frac{4mg}{5}$$

$$K\vec{x} + \vec{T} + m\vec{g} = 0$$

$K\vec{x}$ does not change suddenly

$$\Rightarrow K\vec{x} + m\vec{g} = m\vec{a} = -\vec{T}$$

$$ma = \frac{4mg}{5}$$

$$\Rightarrow a = 8m/s^2$$

3. Ans (0)

When slider is at A, the applied potential difference across the photoelectric cell is a retarding potential difference of 10 V. Energy of incident photon

$$E = \frac{hc}{\lambda} = \frac{1240eV\,nm}{350nm} = 3.5eV$$

Maximum KE of emitted photoelectrons is

$$KE_{max} = \frac{hc}{\lambda} - \phi = 3.5 - 2.2 = 1.3eV$$

stopping potential difference

$$V_s = 1.3V$$

(a) The applied potential difference is

$$V\ (=10V) > V_8\ (=1.3V)$$

Hence there is no photocurrent and reading of ammeter is zero.

4. Ans (8)

By the conservation of energy

$$\frac{1}{2}mV_0^2 = mgh + \frac{1}{2}mv^2 \ldots\ldots\ldots\ldots (1)$$

By the conservation of angular momentum

$$\frac{mV_0 r}{2} = mV\,R \ldots\ldots (2)$$

From (1) & (2)

$$V_0 = \sqrt{\frac{8gh}{3}}$$

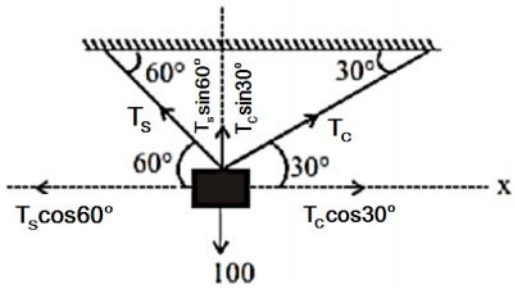
$k = 8$

5. Ans (2)

Let T_S = tension in steel wire

T_C = Tension in copper wire

In x direction



$T_C \cos30° = T_S \cos60°$

$T_C \times \frac{\sqrt{3}}{2} = T_S \times \frac{1}{2}$

$\sqrt{3}T_C = T_S$ ..... (i)

In y direction

$T_C \sin30° + T_S \sin 60° = 100$

$\frac{T_C}{2} + \frac{T_S\sqrt{3}}{2} = 100$ ..... (ii)

Solving equation (i) & (ii)

$T_C = 50N$

$T_S = 50\sqrt{3}N$

We know $\Delta L = \frac{FL}{AY}$

$= \frac{\Delta L_C}{\Delta L_S} = \frac{T_C L_C}{A_C Y_C} \times \frac{A_S Y_S}{T_S L_S}$

On solving above equation

$\frac{\Delta L_C}{\Delta L_S} = 2$

6. Ans (9)

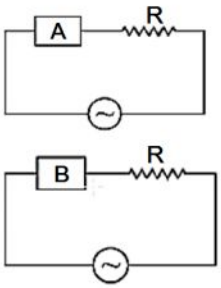
Since water resistance $\left(\vec{F} = -k\vec{v}_{pr}\right)$ is proportional to the velocity of plank relative to the water, impulse of water resistance equals to product of the proportionality constant k and displacement of the plank relative to the water. And this impulse stops the block- plank system at the other bank, therefore equals to the initial momentum of the block.

$mu_{min} = k\int vdt$

$\Rightarrow mu_{min} = k(b - 1)$

$u_{min} = \frac{k(b-\ell)}{m} = 9.0m/s$

7. Ans (2)



$\phi_1 = -30°$

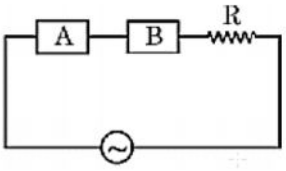
$\phi_2 = +60°$

$\tan\phi = \frac{X_L}{R}$

$\tan\phi_2 = \frac{X_C}{R}$

$X_L = \frac{R}{\sqrt{3}}$

$X_C = R\sqrt{3}$



$\tan\theta = \frac{(X_C - X_L)}{R} = \frac{\frac{R\sqrt{3}-r}{\sqrt{3}}}{R} = \frac{2}{\sqrt{3}}$

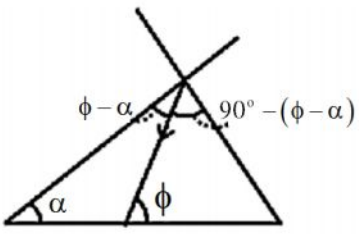
8. Ans (2)

$a_1\cos[90° - (\phi - \alpha)] = a_2\sin\alpha$

$a_1\sin(\phi-\alpha) = a_2\sin\alpha$ ..... (i)

Conservation has no acceleration x direction

$\Rightarrow Ma_2 = ma_1\sin\phi$ ..... (ii)



From (i) and (ii)

$\Rightarrow \frac{m}{M}\sin\phi = \frac{\sin(\phi-\alpha)}{\sin\alpha}$

$\Rightarrow \frac{m}{M} = \frac{\tan\phi - \tan\alpha}{\tan\alpha}$

$\frac{m}{M} = \frac{\tan 60 - \tan 30}{\tan 30} = 2$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

$$HNO_2 + NH_2 - \underset{\underset{O}{\|}}{C} - NH_2 \longrightarrow \underset{(x)}{N_2} + \underset{(y)}{CO_2} + \underset{(z)}{H_2O}$$

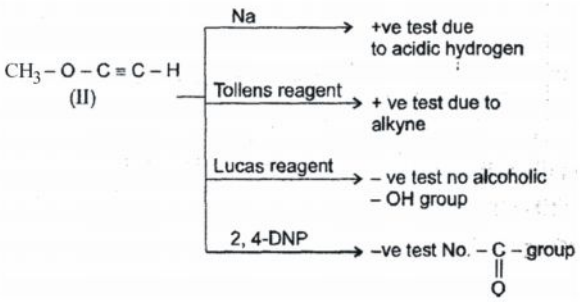
$$\underset{(x)}{Mg + N_2} \longrightarrow \underset{(w)}{Mg_3N_2} \xrightarrow{H_2O} \underset{(p)}{NH_3} \xrightarrow{CuO} \underset{(x)}{N_2}$$

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3$$

Milky solution

$$\underset{(p)}{NH_3} + K_2[HgI_4] / OH^{\odot} \longrightarrow \underset{\text{Brown ppt.}}{HgO \cdot Hg <_{I}^{NH_2}}$$

2. Ans (C)



3. Ans (B)

From the equation,

$$\left[Cu(CN)_4^{3-}\right] = 0.1M \text{ and } \left[CN^{-1}\right] = 0.2M$$

$$\therefore \left[Cu^{+}\right] = \frac{K_{instab..}\left[Cu(CN)_4^{3-}\right]}{[CN^{-}]^4}$$

$$= \frac{6.4 \times 10^{-15} \times 0.1}{(0.2)^4} = 4 \times 10^{-13} M$$

$$\text{Now, } \left[S^{2-}\right] = \frac{K_{SP}(Cu_2S)}{[Cu^{+}]^2} = \frac{2.56 \times 10^{-27}}{\left(4 \times 10^{-13}\right)^2}$$

$$= 1.6 \times 10^{-2} M$$

$$\therefore \left[H^{+}\right] = \sqrt{\frac{K_a \times [H_2S]}{[S^{2-}]}} = \sqrt{\frac{1.6 \times 10^{-21}}{1.6 \times 10^{-2}}}$$

$$= 10^{-10} \text{ M and pH=10.0}$$

4. Ans (D)

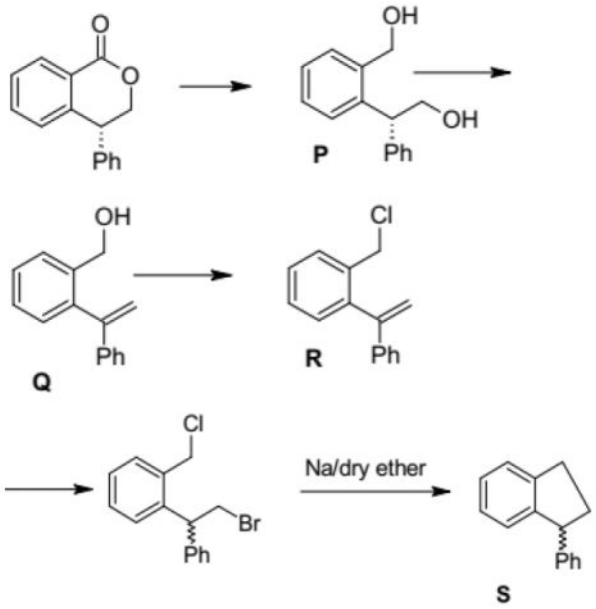
$$n=(1+4+9+...+n^2) = \frac{P_f(1+2+3....+n)V}{RT}$$

$$P_f = \frac{P\left[\frac{n(n+1)(2n+1)}{6}\right]}{\frac{n(n+1)}{2}} = \left(\frac{2n+1}{3}\right)P$$

PART-2 : CHEMISTRY

SECTION-I (ii)

6. Ans (A,B)



7. Ans (A,B,C,D)

A) It is deep red colouration due to the formation of $Fe(SCN)_3$.

B) In presence of moisture following reaction will occur:

$CrO_2Cl_2 + H_2O \longrightarrow H_2CrO_4 + HCl$; So test is carried out only in dry test tube.

C) White precipitate of $PbSO_4$ is formed and hence brown ring is not visible.

D) Gives yellow precipitate of CdS.

8. Ans (B,C,D)

A) $2HCO_3^{-} + Mg^{2+} \xrightarrow{\text{cold}} Mg(HCO_3)_2 \text{ (soluble)} \xrightarrow{\Delta}$

$MgCO_3\downarrow + CO_2 + 2H_2O$

B) $Ca(HCO_3)_2 + 2NH_3 \longrightarrow (NH_4)_2CO_3 + CaCO_3\downarrow$

C) True statement

D) $CO_3^{2-} + 2Hg^{2+} + 3H_2O \longrightarrow$

$Hg_4O_3CO_3\downarrow$ (reddish brown) $+ 6H^{+}$

(basic mercury (II) carbonate)

9. Ans (A,B,D)

$I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

nf = 2 nf =1

No. of Meq in Part-I = No. of meq of hypo

$= 15 \times 0.4$

No. of Meq I_2 = 6 meq = 3 millimoles

Part-II

$3I_2 + 6NaOH \longrightarrow 5NaI + NaIO_3 + 3H_2O$

Millimoles of I_2 with NaOH

$= \dfrac{(30) - (2 \times 3)}{2} = 12$ millimoles

Total number of millimoles of I_2 = 12 + 3

=15 millimoles

Initial molarity = $\dfrac{15\,\text{mmol}}{150\,\text{ml}} = 0.1M$

PART-2 : CHEMISTRY

SECTION-III

2. Ans (4)

SO_2, SO_3, H_3PO_4 and $HClO_4$, has $p\pi - d\pi$ bonding

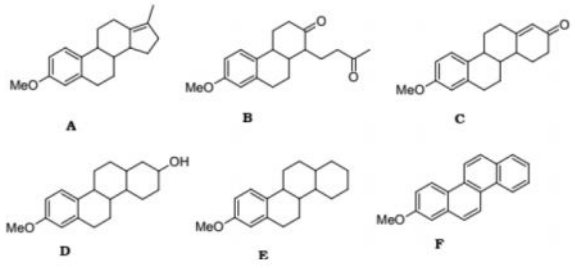
3. Ans (4)

$Hg^{2+} + Co(SCN)_2 \longrightarrow Co[Hg(SCN)_4]$ (deep blue)

4. Ans (8)

A) ClF_3 ; x = 11 B) UF_6 ; y = 18 C) ClF; z = 6

5. Ans (7)



6. Ans (4)

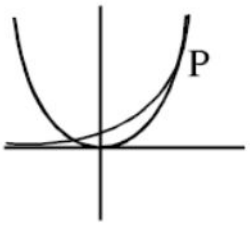
b, c, e, j only

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

Since both curves touches each other



$\Rightarrow m_1 = m_2$

$4(x - a) = 2e^{2x}$

$4(x - a) = 2.2(x - a)^2$

$\Rightarrow (x - a)(x - a - 1) = 0$

$\Rightarrow x = a$ or $x = a + 1$

If $x = a$ then (I^{st}) curve $\Rightarrow y = 0$

& 2nd curve $\Rightarrow y = e^{2a}$ but $e^{2a} \neq 0\ \forall\ a \in R$

$\therefore x \neq a$

$\Rightarrow x = a + 1$

$\therefore 1^{st}$ curve $\Rightarrow y = 2 \therefore (x, y) = (a + 1, 2)$

(x, y) lies on 2nd curve also, $\therefore 2 = e^{2a+2}$

$\Rightarrow 2a + 2 = \ell n2$

$a = \dfrac{1}{2}(\ell n2 - 2)$

$\Rightarrow -1 < a < 0$

2. Ans (B)

Let $P(t, t^2)$ be the parametric co-ordinates of any point on the parabola $y = x^2$

$\Rightarrow \left.\frac{dy}{dt}\right|_{(t,t^2)} = 2t$

Hence normal at P is $(y - t^2) = \frac{-1}{2t}(x - t)$

If this line meets the curve again at $Q(t', t'^2)$, then

$.(t'^2 - t^2) = \left(-\frac{1}{2t}\right)(t' - t)$

$\Rightarrow t' + t = -\frac{1}{2t}$

$\Rightarrow t' = -t - \frac{1}{2t}$

If ℓ is length of the normal chord then $z = (\ell)^2$

$= PQ^2$

$\Rightarrow z = (t - t')^2 [1 + (t + t')^2]$

$\Rightarrow z = \left(t + t + \frac{1}{2t}\right)^2 \left(1 + \left(-\frac{1}{2t}\right)^2\right)$

$\Rightarrow z = \frac{\left(4t^2 + 1\right)^3}{16t^4}$

Put $4t^2 = p$ where $p > 0$

$z = \frac{(p + 1)^3}{p^2} = (p + 3) + \frac{3}{p} + \frac{1}{p^2}$

$\Rightarrow \frac{dz}{dp} = \left(1 - \frac{3}{p^2}\right) - \frac{2}{p^3} = 0$

$\Rightarrow (p + 1)^2 (p - 2) = 0$

$\Rightarrow p = 0$ as $p \neq -1$ and $\frac{d^2z}{dp^2} = \frac{6}{p^3} + \frac{6}{p^4} > 0$

Hence z is minimum when $p = 2 = 4t^2$

3. Ans (D)

$I(k) = \int_0^\infty \frac{\ell nx}{x^2 + kx + k^2} dx$

Put $x = \frac{k}{t}$ we get

$I(k) = \frac{1}{k} \int_0^\infty \frac{\ell nk - \ell nt}{t^2 + t + 1}$

$\Rightarrow I(k) = \frac{\ell nk}{k} \int_0^\infty \frac{dt}{t^2 + t + 1} - \frac{I(1)}{k}$

Now $I(1) = -I(1)$

$I(k) > \frac{\ell nk}{k} \int_0^1 \frac{dt}{t^2 + t + 1} (k > 1)$

4. Ans (D)

Because $|z| \leq 1$ and $|iw| \leq 1$ and $|z - iw| = 2$, so z, iw are end points of diameter of $x^2 + y^2 = 1$. Similarly z and $i\bar{w}$ are also end points of same diameter (because $|z - i\bar{w}| = 2$)

$\Rightarrow iw = i\bar{w} \Rightarrow$ iw lies on real axis

$\Rightarrow$ z lies on real axis

$\Rightarrow$ either $z = 1$, $iw = -1$ or $z = -1$, $iw = 1$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,C)

Let $t_r = \left(\frac{1}{r(r+1)}\right)^3 = \left(\frac{1}{r} - \frac{1}{r+1}\right)$

$= \frac{1}{r^3} - \frac{1}{(r+1)^3} - \frac{3}{r(r+1)}\left(\frac{1}{r(r+1)}\right)$

$= \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3\left(\frac{(r+1) - r}{r(r+1)}\right)^2,$

$t_r = \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3\left(\frac{1}{r^2} + \frac{1}{(r+1)^2}\right) + 6\left(\frac{1}{r} - \frac{1}{r+1}\right)$

$\sum_{r=1}^{n} t_r = \lim_{n\to\infty}\left(1 - \frac{1}{(n+1)^3}\right) - 3\left(\frac{\pi^2}{6} + \frac{\pi^2}{6} - 1\right) + 6\left(1 - \frac{1}{n+1}\right)$

$\lim_{n\to\infty} \sum t_r = 10 - \pi^2$

6. Ans (A,B,C)

Since $\sec^2\theta > 1$

$\Rightarrow [(n+1)\sec^2\theta] > [n\sec^2\theta]$

Hence, f and g are both one-one

Let $k < n\sec^2\theta < k+1$ and

$k < m\,\text{cosec}^2\theta < k+1$

$k\cos^2\theta < n < (k+1)\cos^2\theta$ and

$k\sin^2\theta < m < (k+1)\sin^2\theta$

Adding given k <integer < k + 1

$\Rightarrow A \cap B = \phi$

Suppose $k \notin A$

$\Rightarrow n\sec^2\theta < k$ and $(k+1) < (n+1)\sec^2\theta$

$\Rightarrow n < k\cos^2\theta$ and $n + 1 > (k+1)\cos^2\theta$

$\Rightarrow k < (k-n)\,\text{cosec}^2\theta,\ k+1 > (k-n)\,\text{cosec}^2\theta$

$\Rightarrow [(k-n)\cos ec^2\theta] = k$

$\Rightarrow k \in B$

7. Ans (B,C)

Required probability =

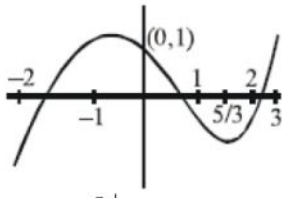
$$\frac{P\left(P_1 \text{ reaches final } \cap\ P_2 \text{ reaches semifinal but not in final}\right)}{P\left(P_2 \text{ reaches semifinal but not in final}\right)}$$

$$= \frac{P\left(P_1 \text{ and } P_2 \text{ reaches semifinal } \cap\ P_1 \text{ wins semifinal but } P_2 \text{ loses semifinal}\right)}{P\left(P_2 \text{ reaches semifinal and } P_2 \text{ loses semifinal}\right)}$$

$$= \frac{\frac{{}^{14}C_2}{{}^{16}C_4} \times \left(\frac{1}{3}\right)\left(\frac{1}{2} + \frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\right)}{\frac{{}^{15}C_3}{{}^{16}C_4} \times \frac{1}{2}} = \frac{2}{15}$$

8. Ans (B,D)

$$= \left(\frac{14\times13\times12}{16\times15\times14\times13}\right)\left(\frac{8}{3}\right) = \left(\frac{2}{15}\right)$$



Clearly, $\begin{vmatrix} x & x & 5 \\ 1 & x & 0 \\ 1 & 1/5 & x \end{vmatrix} = 0 \Rightarrow x^3 - x^2 - 5x + 1 = 0$

Assume $f(x) = x^3 - x^2 - 5x + 1$

$\therefore f'(x) = 3x^2 - 2x - 5 = 0 \Rightarrow x = -1\ \&\ \frac{5}{3}$

Clearly $[\alpha] + [\beta] + [\gamma] = -2 + 0 + 2 = 0$

Now $x^3 - x^2 - 5x + 1 = (x-\alpha)(x-\beta)(x-\gamma)$

Put $x = -1$, we get

$(1+\alpha)(1+\beta)(1+\gamma)$

$= -(-1 - 1 + 5 + 1) = -4$

Now, volume of parallelepiped

$= |x^3 - x^2 - 5x + 1| \times \in [-2, 3]$

Maximum value will occur at $x = \frac{5}{3}$ and is equal to

$$\left|\frac{125}{27} - \frac{25}{9} - \frac{25}{3} + 1\right| = \left|\frac{125 - 75 - 225 + 27}{25}\right| = \frac{148}{27}$$

9. Ans (B,C,D)

Orthocentre of ΔPQR lies on hyperbola

$\Rightarrow \lambda = 3$

Incentre of $\Delta ABC$is orthocenter of ΔPQR

Equation of AB is $x + y - \frac{26}{5} = 0$

10. Ans (A,C,D)

Obviously $f(1+\alpha) = f(1-\alpha)\ \forall\ \alpha \in (0,1)$

Domain of f is (0, 2)

Hence (A) is true

Maxima of $\ln(2x - x^2)$ as well as that of

$\sin x\left(\frac{\pi x}{2}\right)$ occur at x = 1.

Hence (C) is true

As $x \to 0^+$ or $x \to 2^-$, $f(x) \to -\infty$

Hence (D) is true.

PART-3 : MATHEMATICS

SECTION-III

1. Ans (3)

$f(2) = 8 - (r-1)\,4 - 22 + 4r = -ve$

$f(-2) = -8 - (r-1)\,4 + 22 + 4r = +ve$

$\Rightarrow$ One of the integral root can be, -1 or 0 or 1

Case - I : $x = -1$ is one root $\Rightarrow r = -\frac{11}{3}$ rejected

Case - II : $x = 0$ is one root $\Rightarrow r = 0$ but for $r = 0$, $f(x) = x^3 - x^2 - 11x$ which does not have all integral roots

$\Rightarrow r = 0$ (rejected)

Case - III : $x = 1$ is one root $\Rightarrow r = 3$ and at $r = 3$ all roots are integral

(Equation is $x^3 - 2x^2 - 11x + 12 = 0$

$\Rightarrow (x-1)(x-4)(x+3) = 0$)

2. Ans (4)

$$4I + 3J = \int_0^{\frac{\pi}{2}} 1\,dx = \frac{\pi}{2}$$

$$3I - 4J = \int_0^{\frac{\pi}{2}} \frac{3\cos x - 4\sin x}{3\sin x + 4\cos x + 25}\,dx$$

$$= [\ln(3\sin x + 4\cos x + 25)]_0^{\frac{\pi}{2}} = \ln 28 - \ln 29$$

$$3I - 4J = \ln\left(\frac{28}{29}\right) \qquad ...(i)$$

$$4I + 3J = \frac{\pi}{2} \qquad ...(ii)$$

$$11I + 2J = \pi + \ln\left(\frac{28}{29}\right) \equiv 2\pi + b\ln\left(\frac{c}{d}\right)$$

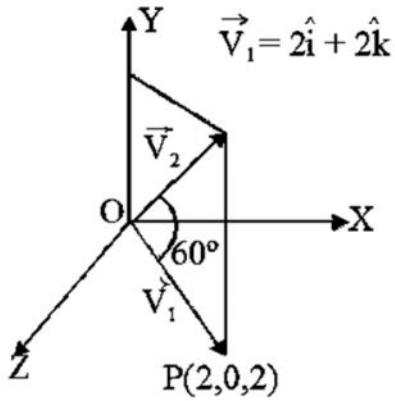
$\Rightarrow d = 1$ and $b = 1$, $c = 28$, $d = 29$

$\therefore a + b + c + d - 55 = 59 - 55 = 4$

3. Ans (4)

Equation of plane perpendicular to XOZ plane is

P : $x - z = 0$, $\vec{V}_2$ is coplanar with $\vec{V}_1$ and j.



$\Rightarrow \vec{V}_2 = x\vec{V}_1 + y\hat{j}$

$\Rightarrow \vec{V}_2 = x\left(2\hat{i} + 2\hat{k}\right) + y\hat{j}$

$\therefore \left|\vec{V}_2\right| = \left|\vec{V}_1\right| = 2\sqrt{2}$

$\therefore 4x^2 + y^2 + 4x^2 = 8$ ...(i)

Now, $\vec{V}_2 . \vec{V}_1 = \left|\vec{V}_2\right|\left|\vec{V}\right|_1 \cos\frac{\pi}{3} = 4$

$4 = \left(x\left(2\hat{i} + 2\hat{k}\right) + y\hat{j}\right) . \left(2\hat{i} + 2\hat{k}\right)$

$4 = 8x \Rightarrow x = \frac{1}{2} \Rightarrow y = \pm\sqrt{6}$

$\Rightarrow \vec{V}_2 = \hat{i} + \sqrt{6}\hat{j} + \hat{k}\left(\because \vec{V}_2 . \hat{j} > 0\right)$

Now, $\vec{V}_3 = \vec{V}_1 + x\hat{i}$

$\vec{V}_3 . \vec{V}_1 = 0 \Rightarrow 8 + 2x = 0 \Rightarrow x = -4$

$\Rightarrow \vec{V}_3 = -2\hat{i} + 2\hat{k}$

$\Rightarrow$ Volume of tetrahedron

$$V = \left|\frac{1}{6}\begin{vmatrix} 2 & 0 & 2 \\ 1 & \sqrt{6} & 1 \\ -2 & 0 & 2 \end{vmatrix}\right|$$

$\Rightarrow V = \frac{1}{6}8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}} = 4$

4. Ans (3)

$h'(x) = f(x)$

$h''(x) = f'(x)$

$h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = 1 = g(1)$

$\Rightarrow f'(g(x))\, g'(x) = 1$

$\Rightarrow f'(g(1))g'(1) = 1 \Rightarrow f'(1)g'(1) = 1 \Rightarrow g'(1) = 1$

$G(x) = x^2 g(x) - xh\,(g(x))$

$G'(x) = 2xg(x) + x^2 g'(x) - h(g(x)) - xh'(g(x))g'(x)$

$= 2xg(x) + x^2 g'(x) - h(g(x)) - x^2 g'(x)$

$= 2xg(x) - h(g(x))$

$G''(x) = 2g(x) + 2xg'(x) - h'(g(x))g'(x)$

$= 2g(x) + xg'(x)$

$\therefore G''(1) = 2g(1) + g'(1) = 2 + 1 = 3$

5. Ans (0)

$(x + 1)^3 + (y + 1)^3 = 16$

$y = -1 + \left(16 - (x + 1)^3\right)^{\frac{1}{3}} = f(x)$

$x = -1 + \left(16 - (y + 1)^3\right)^{\frac{1}{3}}$

$\therefore f^{-1}(x) = -1 + \left(16 - (x + 1)^3\right)^{1/3} = g(x)$

$\Rightarrow f(x) = g(x)$

$f\left(-1 + \sqrt[3]{15}\right) = g\left(-1 + \sqrt[3]{15}\right) = 0$

$(f(x + g(x)).g(x + f(x))) = (f(x + f(x)))^2$

$\frac{d}{dx}(f(x+f(x)))^2 = 2f(x + f(x))$

$f'(x + f(x)).\,(1 + f'(x))$

$\frac{d}{dx}(f(x + f(x)))^2$ at $x = -1 + \sqrt[3]{15} = 0$

$\because g\left(-1 + \sqrt[3]{15}\right) = f\left(-1 + \sqrt[3]{15}\right) = 0$

6. Ans (1)

$P_1 : x + y - z = 1$

$P_2 : (m - 1)x + (3m - 7)y - 5z = -3$

$P_3 : (m - 2)x + (2m - 5)y = -4$

$\because$ all the three planes intersects at a line

$$\because \begin{vmatrix} 1 & 1 & -1 \\ m-1 & 3m-7 & -5 \\ m-2 & 2m-5 & 0 \end{vmatrix}$$

$\Rightarrow m = 3, -2$

At $m = 3$

$$\left.\begin{array}{l} P_1 : x + y - z = 1 \\ P_2 : 2x + 2y - 5z = -3 \\ P_3 : x + y = -4 \end{array}\right\} \text{System does not have any solution}$$

but at $m = -2$

$P_1 : x + y - z = 1$

$P_2 : 3x + 13y + 5z = 3$

$P_3 : 4x + 9y = 4$

$\Rightarrow 5P_1 + P_2 = 2P_3$

$\Rightarrow$ Direction ratio of line of intersection are given by

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 3 & 13 & 5 \end{vmatrix} = 18\hat{i} - 8\hat{j} + 10\hat{k}$$

$\equiv < 9, -4, 5 >$

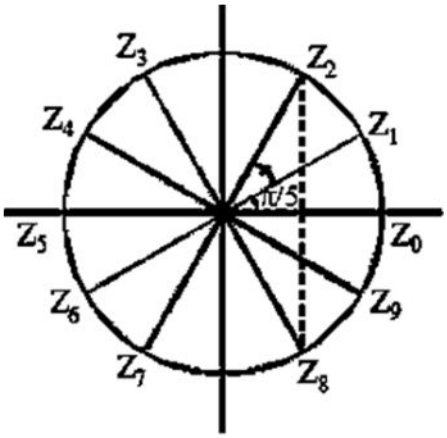
$\Rightarrow a = 9\lambda, b = -4\lambda, c = 5\lambda$

$\therefore \frac{a + b}{c} = 1$

7. Ans (8)

$z = \cos\frac{\pi}{5} + 2i\sin\frac{\pi}{10}\cos\frac{\pi}{10}$

$\Rightarrow z = \cos\frac{\pi}{5} + i\sin\frac{\pi}{5} = e^{i\frac{\pi}{5}}$



$\because Z_p = z^p = e^{ip\frac{\pi}{5}}$, p = 0,1,2,3,....9

$(z_2,z_0)(z_3,z_1)(z_4,z_2)(z_5,z_3)(z_6,z_4)(z_7,z_6)(z_8,z_6)$

(z_9,z_7)

$\Rightarrow$ 8 ordered pairs

8. Ans (7)

$\frac{dy}{dx} = (x+1) + \frac{y-3}{x+1}$

$y = x^2 + x + 3 + c(x+1);$

$\because y_{(x=2)} = 0$ so C = – 3

$f(x) = x^2 - 2x$

$\text{Area} = \int_0^2 2x - x^2 dx = \frac{4}{3}$; m + n = 7